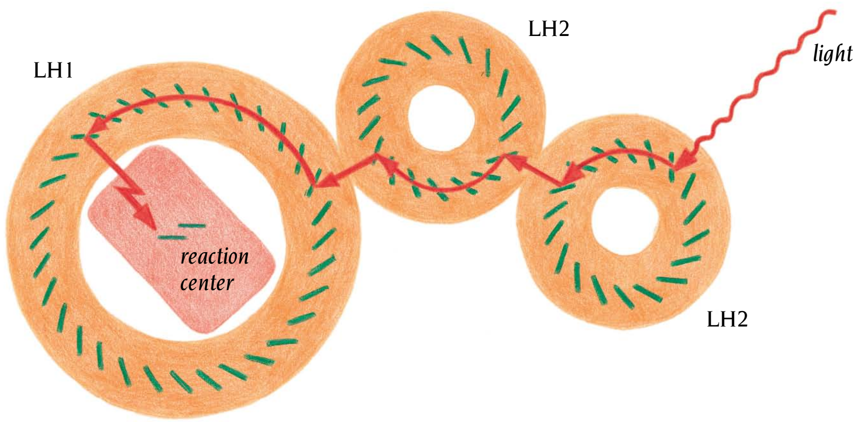

Figure 12.22 Schematic diagram showing the flow of excitation energy in the bacterial photosynthetic apparatus. The energy of a photon absorbed by LH2 spreads rapidly through the periplasmic ring of bacteriochlorophyll molecules (green). Where two complexes touch in the membrane , the energy can be transmitted to an adjacent LH2 ring. From there it passes by the same mechanism to LH1 and is finally transmitted to the special chlorophyll pair in the reaction center. (Adapted from W. Kühlbrandt, *Structure* 3: 521–525, 1995.)

Modeling of the reaction center inside the hole of LH1 shows that the primary photon acceptor—the special pair of chlorophyll molecules—is located at the same level in the membrane, about 10 Å from the periplasmic side, as the 850-nm chlorophyll molecules in LH2, and by analogy the 875-nm chlorophyll molecules of LH1. Furthermore, the orientation of these chlorophyll molecules is such that very rapid energy transfer can take place within a plane parallel to the membrane surface. The position and orientation of the chlorophyll molecules in these rings are thus optimal for efficient energy transfer to the reaction center.

Energy transfer and energy conversion in photosynthetic systems occur virtually without energy loss, whereas solid-state solar cells operate at efficiencies of around 20%. By learning some of the lessons implicit in the arrangement and the reactions of the pigment molecules in the bacterial photosynthetic membrane, and applying these principles to the design of new types of solar cells, it may one day become possible to devise a system of clean, environmentally compatible energy production that operates efficiently at low light intensities.

Transmembrane α helices can be predicted from amino acid sequences

We have seen in previous chapters that only short continuous regions of the polypeptide chains contribute to the hydrophobic interior of water-soluble globular proteins. In such proteins α helices are generally arranged so that one side of the helix is hydrophobic and faces the interior while the other side is hydrophilic and at the surface of the protein as discussed in Chapter 3. Beta strands are usually short, with the residues alternating between the hydrophobic interior and hydrophilic surface. Loop regions between these secondary structure elements are usually very hydrophilic. Therefore, in soluble globular proteins, regions of more than 10 consecutive hydrophobic amino acids in the sequence are rarely encountered.

In contrast, the transmembrane helices observed in the reaction center are embedded in a hydrophobic surrounding and are built up from continuous regions of predominantly hydrophobic amino acids. To span the lipid bilayer, a minimum of about 20 amino acids are required. In the photosynthetic reaction center these α helices each comprise about 25 to 30 residues, some of which extend outside the hydrophobic part of the membrane. From the amino acid sequences of the polypeptide chains, the regions that comprise the transmembrane helices can be predicted with reasonable confidence.

Naively, one might assume that it should be possible to scan the sequence and pick out regions with about 20 consecutive hydrophobic amino acids. However, no such regions occur in the reaction center proteins. Just as in soluble proteins there are hydrophobic side chains at the

Table 12.1 Hydrophobicity scales

Amino acid	Phe	Met	Ile	Leu	Val	Cys	Trp	Ala	Thr	Gly	Ser	Pro	Tyr	His	Gln	Asn	Glu	Lys	Asp	Arg
A	2.8	1.9	4.5	3.8	4.2	2.5	-0.9	1.8	-0.7	-0.4	-0.8	-1.6	-1.3	-3.2	-3.5	-3.5	-3.5	-3.9	-3.5	-4.5
B	3.7	3.4	3.1	2.8	2.6	2.0	1.9	1.6	1.2	1.0	0.6	-0.2	-0.7	-3.0	-4.1	-4.8	-8.2	-8.8	-9.2	-12.3

Row A is from J. Kyte and R.F. Doolittle; row B, from D.A. Engelman, T.A. Steitz, and A. Goldman.

hydrophilic surface of the molecule, in the transmembrane helices of the reaction center there are hydrophilic side chains (which are often important for function) among the hydrophobic. However, hydrophobic residues are in a clear majority in transmembrane helices, and such residues occur less frequently in other continuous regions of the polypeptide chain. Therefore, we need some method to measure the amount of hydrophobicity in a segment of the amino acid sequence in order to be able to predict whether or not it is likely to be a transmembrane helix.

Hydrophobicity scales measure the degree of hydrophobicity of different amino acid side chains

Each amino acid side chain within a transmembrane helix has a different hydrophobicity. It is easy to state that side chains such as Val, Met, and Leu are the most hydrophobic and that charged residues such as Arg and Asp are at the other end of the scale. However, to order all side chains according to hydrophobicity and to assign actual numbers that represent their degree of hydrophobicity is not trivial. Many such **hydrophobicity scales** have been developed over the past decade on the basis of solubility measurements of the amino acids in different solvents, vapor pressures of side-chain analogs, analysis of side-chain distributions within soluble proteins, and theoretical energy calculations. In Table 12.1 two of these hydrophobicity scales are listed. The most frequently used scale, which was introduced by J. Kyte and R.F. Doolittle at University of California, San Diego, is based on experimental data. A more refined scale was developed by D.A. Engelman, T.A. Steitz, and A. Goldman at Yale University. They used a semitheoretical approach to calculating the hydrophobicity, taking into account the fact that the side chains are attached to an α-helical framework.

Hydropathy plots identify transmembrane helices

These hydrophobicity scales are frequently used to identify those segments of the amino acid sequence of a protein that have hydrophobic properties consistent with a transmembrane helix. For each position in the sequence, a hydropathy index is calculated. The **hydropathy index** is the mean value of the hydrophobicity of the amino acids within a "window," usually 19 residues long, around each position. In transmembrane helices the hydropathy index is high for a number of consecutive positions in the sequence. Charged amino acids are usually absent in the middle region of transmembrane helices because it would cost too much energy to have a charged residue in the hydrophobic lipid environment. It might be possible, however, to have two residues of opposite charge close together inside the lipid membrane because they neutralize each other. Such charge neutralization has been observed in the hydrophobic interior of soluble globular proteins.

When the hydropathy indices are plotted against residue numbers, the resulting curves, called hydropathy plots, identify possible transmembrane helices as broad peaks with high positive values. Such hydropathy plots are shown in Figure 12.23 for the L and M chains of the reaction center.

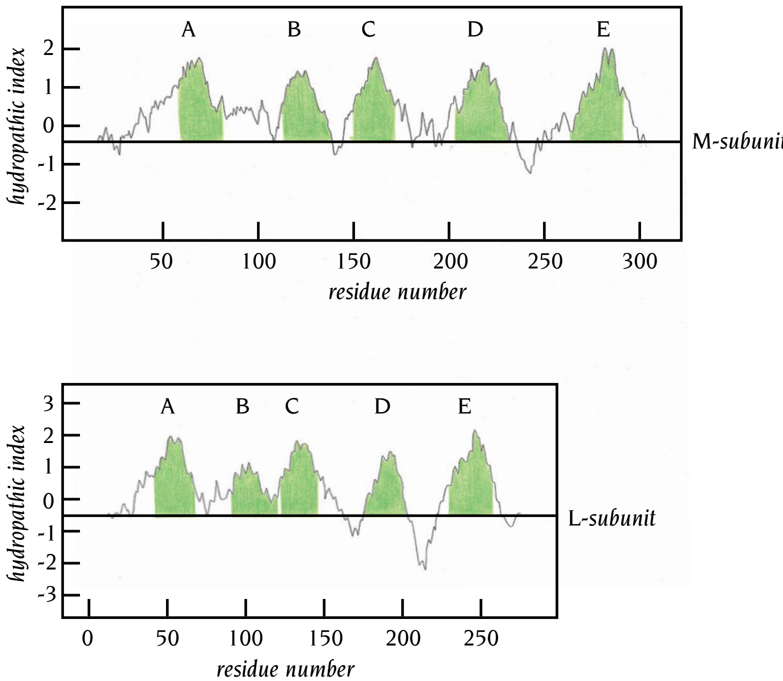

Figure 12.23 Hydropathy plots for the polypeptide chains L and M of the reaction center of *Rhodobacter sphaeroides*. A window of 19 amino acids was used with the hydrophobicity scales of Kyte and Doolittle. The hydropathy index is plotted against the tenth amino acid of the window. The positions of the transmembrane helices as found by subsequent x-ray analysis by the group of G. Feher, La Jolla, California, are indicated by the green regions.

Reaction center hydropathy plots agree with crystal structural data

The hydropathy plots in Figure 12.23 were calculated and published several years before the x-ray structure of the reaction center was known. It is therefore of considerable interest to compare the predicted positions of the transmembrane-spanning helices with those actually observed in the x-ray structure. These observed positions are indicated in green in Figure 12.23.

It is immediately apparent that these plots correctly predict the number of transmembrane helices, five each in the L and M chains, and also their approximate positions in the polypeptide chain. This gives us confidence in the hydropathy plot method. Transmembrane helices are the only secondary structure elements that can be predicted from novel amino acid sequences with a high degree of confidence using current knowledge and methods. The exact ends of transmembrane helices, however, cannot be predicted, essentially because they are usually inserted within the polar head-groups of the membrane lipids and therefore contain charged and polar residues. The transmembrane helices in the reaction center, for example, contain a number of charged residues at their ends (Table 12.2), most of which are at the cytoplasmic side. All of the helices, however, have a segment of at least 19 consecutive amino acids that contain no charged side chains. The majority of residues in these segments are hydrophobic, but there are a number of polar residues, such as Ser, Thr, Tyr, and Trp, among them. The presence of histidine residues in the D and E helices of subunits L and M is accounted for by their special function in the reaction center; they are ligands to the magnesium atoms of chlorophyll molecules and to the Fe atom.

Membrane lipids have no specific interaction with protein transmembrane α helices

Comparison of the amino acid sequences of the L and M subunits of the reaction centers from three different bacterial species shows that about 50% of all residues in those two subunits are conserved in all three species. In the transmembrane helices, sequence conservation varies. Residues that are buried and have contacts either with pigments or with other transmembrane helices are about 60% conserved. In contrast, residues that are fully exposed to the membrane lipids are only 16% conserved. Clearly, fewer restrictions are

Table 12.2 Amino acid sequences of the transmembrane helices of the photosynthetic reaction center in *Rhodobacter sphaeroides*

LA	G F F G V A T F F F A A L G I I L I A W S A V L
LB	L K R C I E V E R L A W S V F A G T A C I T I I Q W L G G
LC	H I P F A F A F A I L A Y L T L V L F R P V M
LD	A A S L V L A G H L A L A L A N T F F F S I A I M H A P
LE	G T L G I H R L G L L L S L S A V F F S A L C M I I
MA	S L G V L S L F S G L M W F F T I G I W F W Y Q A
MB	A Q A R L Y T R G W W S W V A V F M F F S A I L W L G G E K L
MC	A W A F L S A I W L W M V L G F I R P I L M
MD	V A L I T A G H M A F L L A S G Y L F A I S L G H F P
ME	M E G I H R W A I W M A V L V T L T G G I G I L L
HA	M N E T Q L Y Y I L G A L F I W F S Y I A L S A L

The helices are aligned according to approximate positions within the membrane and with respect to the photosynthetic pigments. LA is the first helix of subunit L, ME is the last helix of subunit M, HA is the only transmembrane helix of subunit H. Charged residues are colored red, polar residues are blue, hydrophobic residues are green, and glycine is yellow. (From T.O. Yeates et al., *Proc. Natl. Acad. Sci. USA* 84: 6438–6442, 1987.)

placed on residues that are exposed to the membrane lipids than to residues having contact with polypeptide chains or pigments. This implies that there are relatively few specific interactions between these transmembrane helices and the fatty acid side chains of the membrane that require the presence of specific residues. This is consistent with the observation that membrane proteins can move within the plane of the membrane, by lateral diffusion, and are not at fixed positions.

Conclusion

The x-ray structure of a bacterial photosynthetic reaction center and the associated light-harvesting complexes has given insight into the mechanism for the primary reaction of photosynthesis: the capture and conversion of light energy to chemical energy by a stable separation of negatively and positively charged molecules. In addition, the structure has provided geometrical constraints for theoretical calculations on the electron flow through pigments across the membrane during this charge separation.

Important novel information has thus been obtained for the specific biological function of those molecules, but disappointingly few general lessons have been learned that are relevant for other membrane-bound proteins with different biological functions. In that respect the situation is similar to the failure of the structure of myoglobin to provide general principles for the construction of soluble protein molecules as described in Chapter 2.

The most important general lesson is that there are hydrophobic transmembrane helices, the positions of which within the amino acid sequence can be predicted with reasonable accuracy. This applies both to the single transmembrane-spanning helix within the H polypeptide chain of the reaction center and the five transmembrane helices of the L and M chains that are connected by loop regions. This does not imply, however, that it is equally simple to predict the positions of transmembrane helices in different classes of proteins by hydropathy plots. The transmembrane helices of ion channels, for example, contain charged residues facing the channel. The latter would give quite different hydropathy plots from a single transmembrane helix in a receptor protein.

The structure of the reaction center also established that membrane-spanning helices can be tilted with respect to the plane of the membrane and that their relative positions within the membrane might be determined by the way they are anchored to the loop regions. Finally, several structures provide examples of how binding pockets for ligands are formed between such transmembrane-spanning helices.

The three-dimensional structure of the bacterial membrane protein, bacteriorhodopsin, was the first to be obtained from electron microscopy of two-dimensional crystals. This method is now being successfully applied to several other membrane-bound proteins.

Unlike the other membrane proteins discussed here, the porins in the outer membrane of Gram-negative bacteria have transmembrane regions that are β strands, not α helices. The β strands, 16 in some porins, 18 in others, are arranged into up and down antiparallel β barrels. These barrels, because they have so many strands in their walls, do not have a tightly packed hydrophobic core. Their center is a channel, which is partially blocked by loop regions between two of the strands, leaving an eyelet about 9 Å long and 8 Å in diameter. The complete functional porin molecule comprises three of these barrels; loop regions from the upper surface of the three barrels are in contact with the extracellular space and form a broad funnel leading to the barrels and via their eyelets to the periplasmic space. The porins in this way form size-restricted channels for the passive diffusion of molecules in and out of the periplasmic space.

Like the photosynthetic reaction center and bacteriorhodopsin, the bacterial K^+ ion channel also has tilted transmembrane helices, two in each of the subunits of the homotetrameric molecule that has fourfold symmetry. These transmembrane helices line the central and inner parts of the channel but do not contribute to the remarkable 10,000-fold selectivity for K^+ ions over Na^+ ions. This crucial property of the channel is achieved through the narrow selectivity filter that is formed by loop regions from the four subunits and lined by main-chain carbonyl oxygen atoms, to which dehydrated K^+ ions bind.

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Signal Transduction 13

Many signal-transducing receptors are plasma membrane proteins that bind specific extracellular molecules, such as growth factors, hormones, or neurotransmitters, and then transmit to the cell's interior a signal that elicits a specific response. These responses are usually cascades of enzymatic reactions giving rise to many different effects within the cell, including changes in gene expression. Interference with receptor-signaling systems can therefore have drastic consequences. For example, oncogene products that simulate the function of receptors or their associated signal transmitters cause the loss of cellular growth control.

The DNA sequences of membrane-bound signal-transducing receptor molecules have shown that many are structurally and evolutionarily related, and they can be grouped into a few families making up three major classes: ion-channel linked receptors homologous to the K^+ channel molecule described in Chapter 12, G protein-linked receptors, and enzyme-linked receptors. These last two classes of receptor molecules have an extracellular domain that recognizes a specific molecular signal, a transmembrane region through which the signal is transmitted, and an intracellular domain that produces a response (Figure 13.1). These proteins provide yet another example of the now familiar story that there are only a limited number of different types of domains and that protein molecules with different functions have evolved either by accumulation of point mutations or by combining domains in different ways, by gene shuffling.

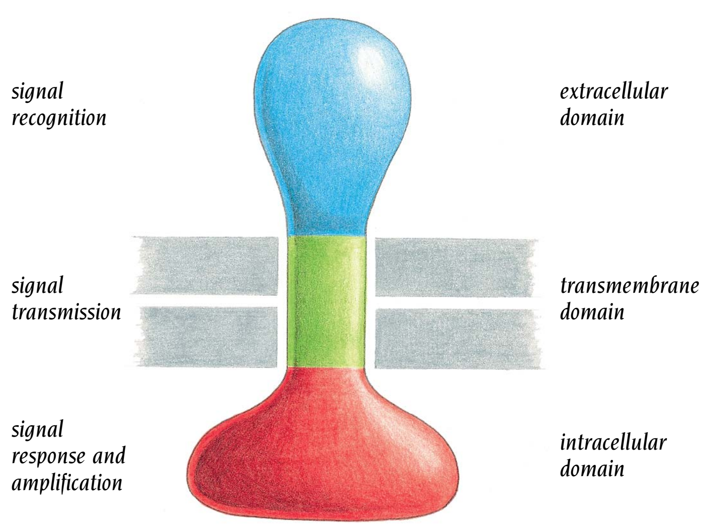


Figure 13.1 The basic organization of a membrane receptor molecule consists of an extracellular domain, a transmembrane region, and an intracellular domain.

No three-dimensional structure is yet available for any complete receptor because they are large and membrane bound, and hence difficult to crystallize, and are too large to study by NMR. However, by recombinant DNA techniques it has been possible to crystallize isolated extracellular and intracellular domains of receptor molecules. In this chapter we will give examples of signal recognition by the binding of growth hormone to the extracellular domain of its receptor, and of the intracellular signal response and amplification by G protein and protein tyrosine kinase-linked receptors. We have no detailed structural knowledge on how signals are transmitted through the membrane, nor on how receptors are linked to ion channel signaling.

G proteins are molecular amplifiers

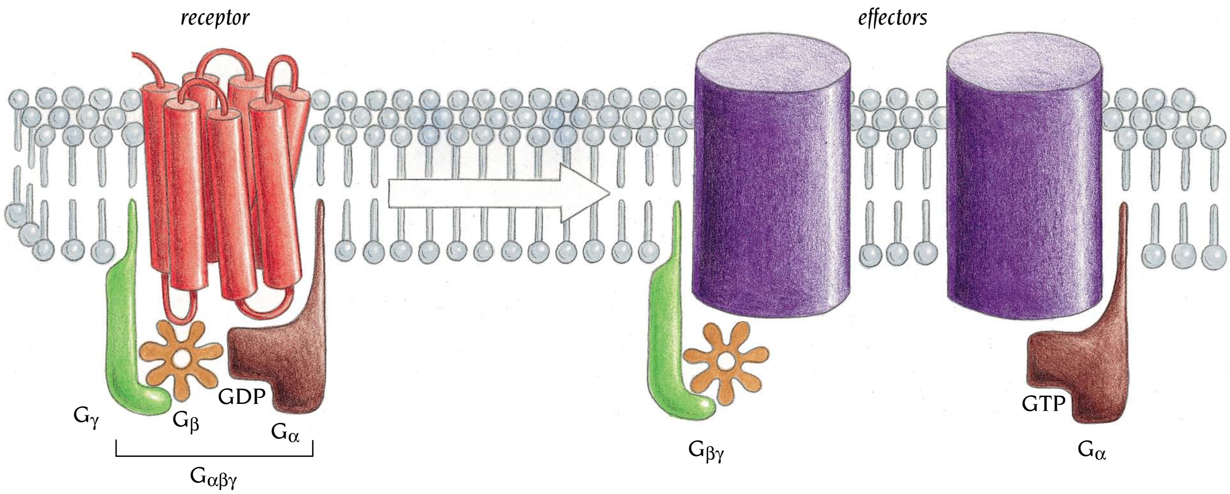
Several important physiological responses, including vision, smell, and stress response, involve large metabolic effects produced from a small number of input signals. The receptors that trigger these responses have two things in common. First, their transmembrane regions contain seven helices each spanning the lipid bilayer of the plasma membrane. Second, the signals transmitted to the intracellular domain of these receptors are amplified, and the amplifiers are members of a common family of proteins with homologous amino acid sequences called **G proteins**. G proteins bind guanine nucleotides (hence the term G protein) and act as molecular switches that are activated by binding GTP and are inactivated when the GTP is hydrolyzed to GDP. The hydrolysis of GTP is catalyzed by the G protein itself, but G proteins on their own are very slow GTPases, and switching off the G protein is normally accelerated by regulatory molecules, known as **RGS** (regulators of GTP hydrolysis), which bind to the active G protein and increase the rate of GTP hydrolysis. When in the active GTP-bound state, the G protein can activate many downstream effectors, greatly amplifying the signal, before RGS binds and the signal is switched off. The structural basis of this important switch mechanism is now understood.

Most G proteins are heterotrimers consisting of one copy each of α (45 kDa), β (35 kDa), and γ (8 kDa) subunits. It is the α subunit that contains the GTPase activity. When the G protein is activated by binding GTP, the heterotrimer dissociates into the α subunit and a $\beta\gamma$ heterodimer. Both the α subunit and the $\beta\gamma$ heterodimer can independently transmit the signal received from the receptor to different effector proteins. The human genome is estimated to contain over 1000 different genes for these receptors with seven transmembrane helices and, in addition, enough genes for different α, β and γ subunits to allow the formation of hundreds of different G proteins. Consequently, there is a very large number of possible combinations of these receptors and G proteins, which provide the potential for a cell to respond to a wide variety of external signals. Table 13.1 gives some examples of physiological responses by G proteins.

Both the α and the γ subunits of G proteins are anchored to the membrane by lipids covalently bound to the N-terminal region of the G_α chain

Table 13.1 Examples of physiological processes mediated by G proteins

Stimulus	*Receptor*	*Effector*	*Physiological response*
Epinephrine	β-adrenergic receptor	adenylate cyclase	glycogen breakdown
Light	rhodopsin	c-GMP phosphodiesterase	visual excitation
IgE–antigen complexes	mast cell IgE receptor	phospholipase C	histamine secretion in all allergic reactions
Acetylcholine	muscarinic receptor	potassium channel	slowing of pacemaker activity that controls the rate of the heartbeat



and the C-terminal region of the G_γ chain (Figure 13.2). The β subunit lacks such lipid modification but remains at the membrane because it is always complexed with the γ subunit, forming the heterodimeric species $G_{\beta\gamma}$. All three species that exist in the cell—G_α, $G_{\beta\gamma}$, and $G_{\alpha\beta\gamma}$—are consequently attached to the cell membrane through the lipid modification of subunits α and γ. On their own both the β and the γ subunits are unstable and easily degraded, whereas the α subunit is stable as a monomer. When GDP is bound to G_α, it forms a heterotrimeric complex with $G_{\beta\gamma}$ but when GTP is bound, G_α remains monomeric. Because they are attached to the cell membrane, G proteins are correctly oriented with respect to each other and to receptors and effector molecules as they move about in two dimensions (see Figure 13.2). Thus, the lipid anchors markedly enhance the search for physiological partners.

In the inactive state, the G protein consists of a G_α–GDP complex combined with a dimeric $G_{\beta\gamma}$ molecule to form a stable heterotrimeric $G_{\alpha\beta\gamma}$, which binds to the cytosolic domain of a receptor (Figure 13.3). When the external domain of a receptor binds its specific ligand (or, as in the case of the visual receptor, rhodopsin, when the ligand is photoisomerized, as discussed in Chapter 12), a signal passes through the membrane causing the cytosolic domain to become activated by a conformational change. The G protein is activated when the changed cytosolic surface of the receptor catalyzes the exchange of bound GDP for GTP. The activated G protein is then released from the receptor and dissociates into the two active components $G_{\beta\gamma}$ and G_α–GTP. These transmit the signal to effector proteins that in turn produce second messenger molecules, such as cyclic AMP, which give rise to physiological responses.

Figure 13.2 Activated G protein receptors, here represented as seven red transmembrane helices, catalyze the exchange of GTP for GDP on the $G_{\alpha\beta\gamma}$ trimer. The then separated G_α–GTP and $G_{\beta\gamma}$ molecules activate various effector molecules. The receptor is embedded in the membrane, and G_α, $G_{\beta\gamma}$ and $G_{\alpha\beta\gamma}$ are attached to the membrane by lipid anchors, and they all therefore move in two dimensions. (Adapted from D. Clapham, *Nature* 379: 297–299, 1996.)

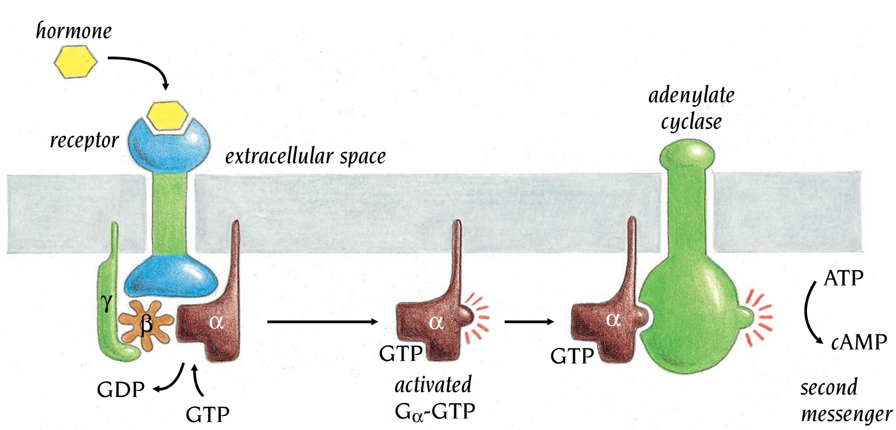


Figure 13.3 G protein–mediated activation of adenylate cyclase by hormone binding. Hormone binding on the extracellular side of a receptor such as the β adrenergic receptor activates a G protein on the cytoplasmic side. The activated form of the G protein (α subunit with bound GTP) dissociates from the β and γ subunits and activates adenylate cyclase to produce cyclic AMP, which is a second messenger that affects a diverse range of cellular processes.

As long as the ligand remains on the extracellular domain of the receptor, its cytoplasmic domain can continue to trigger the production of active G_α–GTP molecules. Each active G_α–GTP, by complexing with an effector molecule, in turn triggers the production of many second messenger molecules. For example, one molecule of the hormone epinephrine bound to the receptor in this way produces a cascade of molecules of the second messenger, cyclic AMP, through the action of G proteins and adenylate cyclase. The GTPase activity of the G_α subunit determines the length of time that the signal remains on. Once hydrolysis to GDP has occurred, the α subunit is inactivated and able to bind to free $G_{\beta\gamma}$, ready for the cycle to recommence.

A failure to turn off GTP-activated G_α has dire consequences. For example, in the disease cholera, cholera toxin produced by the bacterium *Vibrio cholerae* binds to G_α and prevents GTP hydrolysis, resulting in the continued excretion of sodium and water into the gut.

To illustrate the structural basis of signal transduction by G proteins, we shall discuss the following: (i) the structural similarities and differences of G_α and the regulator molecule Ras; (ii) the structural changes that convert resting G_α–GDP to active G_α–GTP; (iii) the mechanism of the intrinsic GTPase activity of G_α and Ras, and how the rate of GTP hydrolysis is changed by binding to regulators; (iv) the structure of the complex of G_α with $G_{\beta\gamma}$, in which the GTPase activity is inhibited and the G protein is prepared for binding to the receptor; (v) the formation of complexes of $G_{\beta\gamma}$ with regulators, which prevents the formation of $G_{\alpha\beta\gamma}$.

Ras proteins and the catalytic domain of G_α have similar three-dimensional structures

The **Ras** proteins, regulators of signal transduction processes leading to cell multiplication and differentiation, are molecular switches activated in response to protein tyrosine kinase receptors. They are GTP-binding proteins with essentially no GTPase activity, and like G_α they are activated as molecular switches by binding GTP and are inactive when complexed with GDP. To be switched off they require the action of a GTPase-activating protein called **GAP**. About 25% of all tumor cells have mutant Ras proteins that cannot be

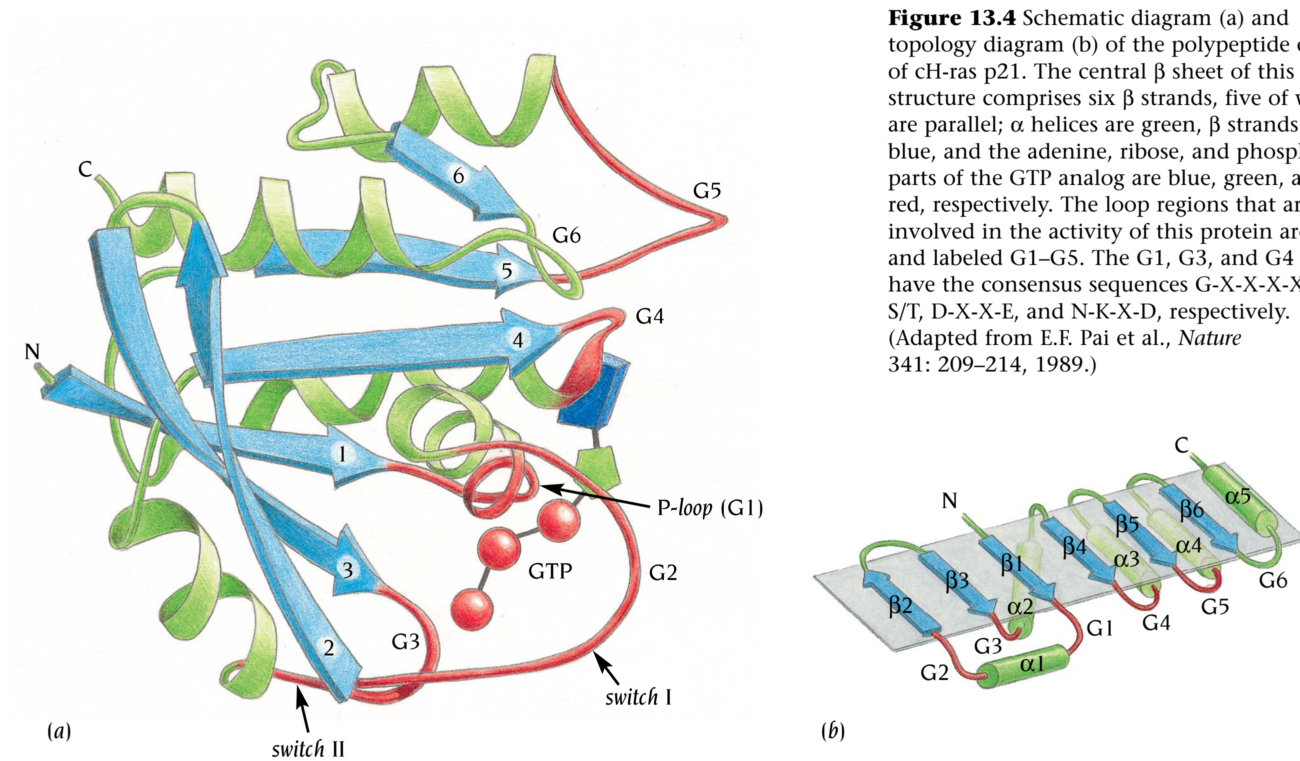


Figure 13.4 Schematic diagram (a) and topology diagram (b) of the polypeptide chain of cH-ras p21. The central β sheet of this α/β structure comprises six β strands, five of which are parallel; α helices are green, β strands are blue, and the adenine, ribose, and phosphate parts of the GTP analog are blue, green, and red, respectively. The loop regions that are involved in the activity of this protein are red and labeled G1–G5. The G1, G3, and G4 loops have the consensus sequences G-X-X-X-X-G-K-S/T, D-X-X-E, and N-K-X-D, respectively. (Adapted from E.F. Pai et al., *Nature* 341: 209–214, 1989.)

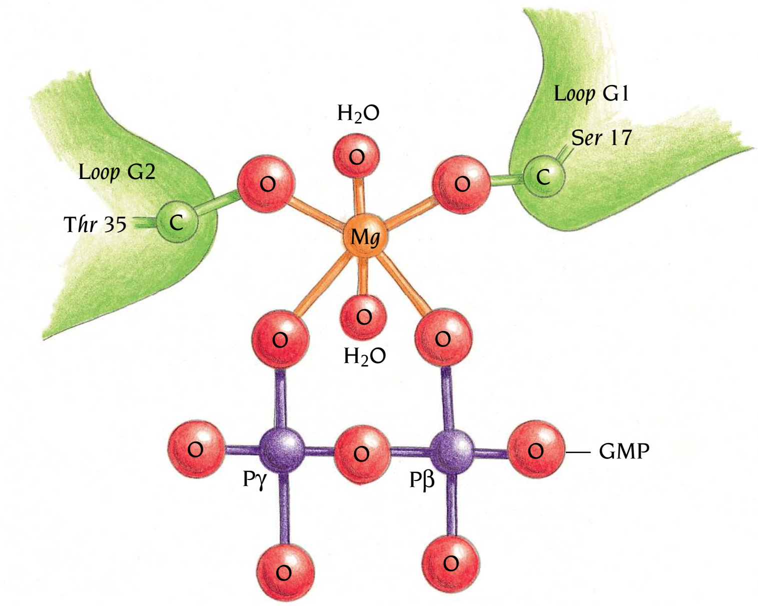


Figure 13.5 A Mg^{2+} atom links GTP to the Ras protein. Mg is coordinated to one oxygen atom each from the β and γ phosphates of GTP as well as to the side chains of Ser 17 and Thr 35 of Ras. Two water molecules complete the octahedral coordination of Mg.

activated to hydrolyze GTP; as a result, these mutant Ras molecules remain bound to GTP and activated, and this leads to uncontrolled cell growth.

The three-dimensional structure of Ras was determined by the groups of Sung-Hou Kim at the University of California, Berkeley, and Ken Holmes at the Max-Planck Institute, Heidelberg. Ras has an α/β-type structure in which the central β sheet comprises six β strands, five of which are parallel (Figure 13.4). There are five α helices positioned on both sides of the β sheet. The GTP is bound in a pocket at the carboxy ends of the β strands in a way similar to the binding of nucleotides to other nucleotide-binding proteins. Loop regions that connect the β strands with the α helices form the binding pocket. The topology of this fold, shown in Figure 13.4b, as well as the mode of nucleotide binding, is exactly the same as that observed earlier by Jens Nyborg in Brian Clark's laboratory at Aarhus University, Denmark, and Frances Jurnak at the University of California, Riverside, in their crystal structures of the elongation factor Tu, which is involved in protein synthesis on the ribosome.

Almost all nucleotide triphosphate hydrolyzing enzymes, including Ras and G_α, require magnesium ions for their catalytic activities. Mg^{2+} is presumably required both for proper positioning of the γ phosphate and for weakening the P–O bond that is split during catalysis, as well as for maintaining the stability of the guanine nucleotide complex. In the structure of Ras and all GTPases bound to a GTP analog, a magnesium ion is coordinated to two phosphate oxygen atoms, one each from the β and γ phosphates. In addition the Mg^{2+} coordinates two water molecules and two protein oxygen ligands, in Ras one from the side chain of Ser 17 in loop G1 and one from Thr 35 in loop G2 (Figure 13.5).

Five of the six loop regions (G1–G5 in Figure 13.4) that are present at the carboxy end of the β sheet in the Ras structure participate in the GTP binding site. Three of these loops, G1 (residues 10–17), G3 (57–60), and G4 (116–119), contain regions of amino acid sequence conserved among small GTP-binding proteins and the G_α subunits of trimeric G proteins.

The first loop region, G1, of Ras, which is also called the diphosphate-binding loop or P-loop, connects the β1 strand with the α1 helix (see Figure 13.4) and is essential for proper positioning of the phosphate groups by binding to the α and β phosphates of the guanine nucleotide. Loop G3, which connects β3 with α2, forms a link between the subsites that bind Mg^{2+} and the γ phosphate of GTP. The consensus amino acid sequences of these two loop regions have been found in other nucleotide-binding proteins such as myosin, described in Chapter 14, and are frequently used as a fingerprint motif to identify from amino acid sequences ATP- and GTP- binding proteins. Loop G2 contains the threonine residue conserved in both Ras and G_α proteins that binds Mg^{2+} and which is important in both structural switching and GTP hydrolysis. Loops G4 and G5 are involved in recognizing and binding the guanine base of the nucleotide. Two regions in the Ras molecule undergo large conformational changes when the molecule switches from the active GTP-bound form to the inactive GDP-bound form. They are called switch regions and are found in loop G2 (switch I) and the region comprising loop G3 and helix α2 (switch II) (Figure 13.4a).

X-ray structures of G_α protein complexes with GDP and GTP analogs that simulate both the ground state and transition states of the GTPase reaction have been determined by the groups of Paul Sigler at Yale University, New Haven and Stephen Sprang at the University of Texas, Dallas. These studies have given a detailed picture of the structural aspects of the mechanism for GTP hydrolysis by G_α. Paul Sigler's group has studied the G_α subunit of **transducin**, a G protein that mediates vision in the retina by transmitting light signals detected by rhodopsin, the photon receptor in the eye, to cyclic GMP phosphodiesterase, the effector. Both groups have obtained essentially similar results with similar conclusions and we will here focus on the G_α subunit of transducin, from which Sigler's group removed the covalently bound lipid at the N-terminus in order to promote crystallization.

The polypeptide chain of transducin G_α is about twice as long as that of Ras and is divided in two domains, a GTPase domain with a structure similar to Ras and an α-helical domain with a unique topology (Figure 13.6). The helical domain is connected to the GTPase domain by two linker regions, one of which follows helix α1 and the other precedes strand β2 in the GTPase domain. The whole helical domain including parts of the linker regions can be regarded as a large insertion in the GTPase domain at a position corresponding to the G2 loop in Ras (compare Figures 13.4 and 13.6). The bound nucleotide, GTP-γS, which is a chemically modified GTP with a sulfur atom instead of oxygen in the γ phosphate to prevent hydrolysis, is completely buried between the two domains. The inaccessibility of the GTP analog in G_α

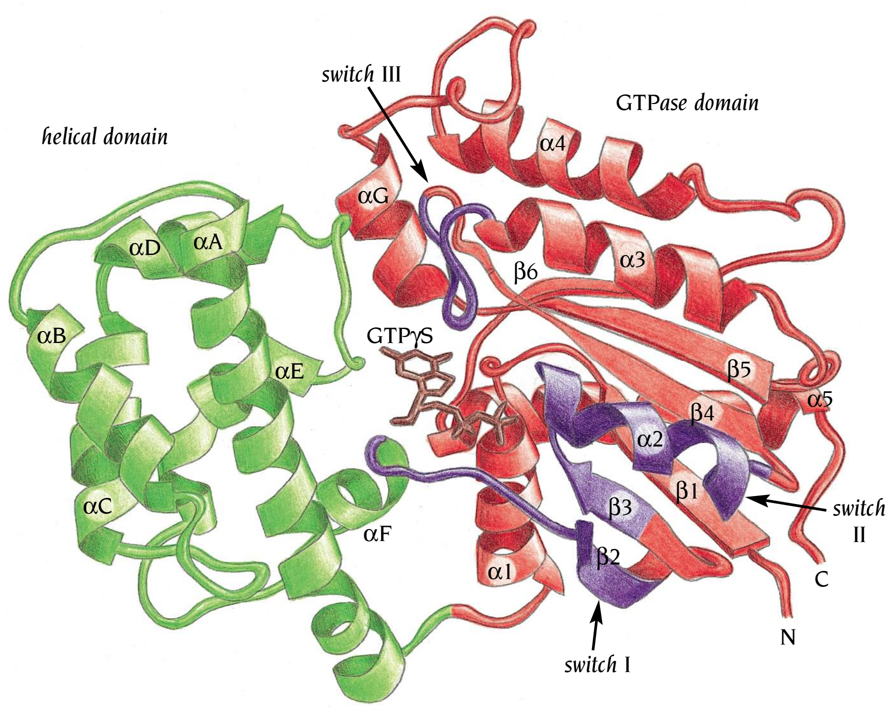


Figure 13.6 Schematic diagram of G_α from transducin with a bound GTP analog. The polypeptide chain is organized into two domains; a catalytic domain (red) with a structure similar to Ras, and a helical domain (green) which is an insert in the loop between α1 and β2. There are three switch regions (violet) that have different conformations in the different catalytic states of G_α. The GTP analog (brown) is bound to the catalytic domain in a cleft between the two domains. (Adapted from J. Noel et al., *Nature* 366: 654–663, 1993.)

suggests that in order to exchange guanine nucleotides there may be a conformational change that opens the cleft.

The helical domain is built up from one long α helix of 28 residues that acts as a scaffold, supporting five smaller helices through hydrophobic interactions. These interactions provide a rigid internal structure consistent with the notion that the whole domain functions as a rigid lid on the nucleotide binding site, and that conformational changes might operate through hinge movements opening and closing the whole lid.

Compared to Ras, G_α subunits are extended at their N-termini by about 30 residues that in free G_α are disordered but, as we will see, obtain a helical conformation in association with $G_{\beta\gamma}$.

G_α is activated by conformational changes of three switch regions

The transduction of signals by G proteins depends on the ability of G_α to cycle between the resting (GDP-bound) conformation primed for interaction with stimulated receptors, and the active (GTP-bound) conformation capable of activating or inhibiting a variety of effector molecules. How can the presence or absence of a phosphate in the bound nucleotide cause such drastic effects? Both Sigler's and Sprang's groups have revealed details of the conformational differences between these two states of G_α by comparing high-resolution structures of G_α complexed with GDP (resting form) and GTP-γS (active form).

The structural differences are essentially localized to three regions in the GTPase domain (see Figure 13.6), two of which are the same as switch regions I and II described for the Ras structure, while the third, switch region III, is another loop region linking β4 with α3, corresponding to G4 in Ras (see Figures 13.4 and 13.6). The structural changes of these regions are considerable, as can be seen in Figure 13.7. The trigger for the conformational change from the GDP form to the GTP form seems to be the formation of hydrogen bonds between the protein and the γ-phosphate of GTP, coupled to the formation of new bonds to the Mg^{2+} ion from protein ligands.

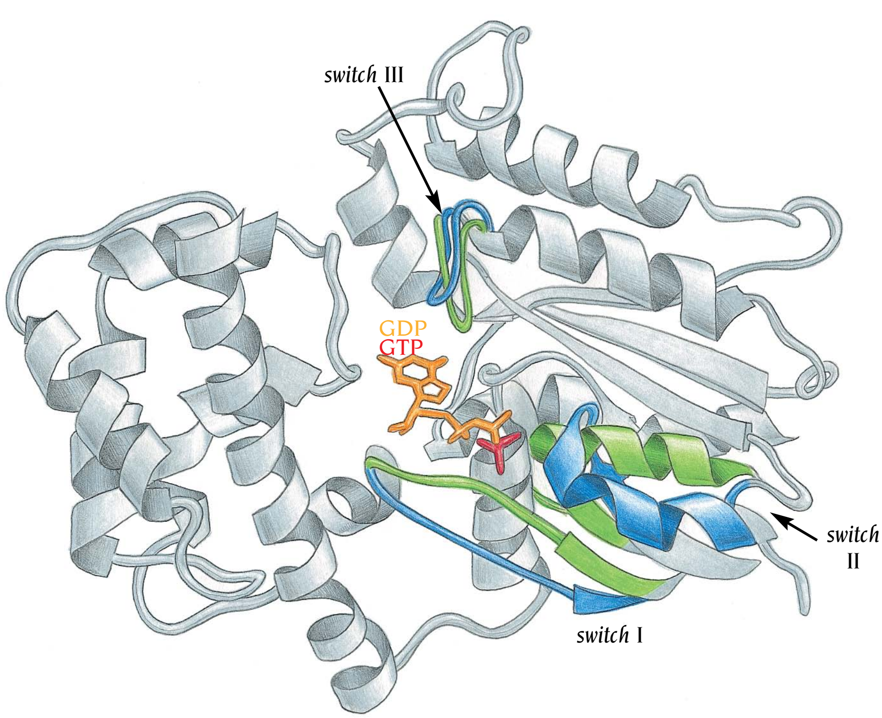


Figure 13.7 Structural differences between the inactive and active forms of G_α. The major part of the structure is unchanged (gray regions) but the three switch regions change conformation. In the active GTP-bound form (green) the switch regions are closer to the catalytic site than they are in the inactive GDP-bound form (blue). (Adapted from D. Lambright et al., *Nature* 369: 621–628, 1994.)

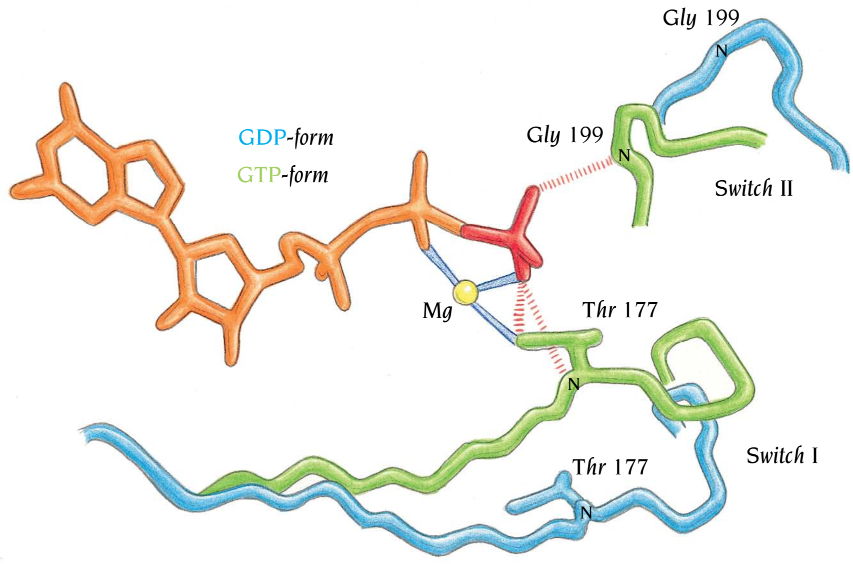


Figure 13.8 Interactions involved in the switch from the inactive GDP- (blue) to the active GTP- (green) bound forms of G_α from transducin. The diagram illustrates the local changes required in the switch I and II regions in order to bring the side chain of Thr 177 and the main chain N of Gly 199 into contact with the γ phosphate of GTP. (Adapted from D. Lambright et al., *Nature* 369: 621–628, 1994.)

In both structures the Mg^{2+} ion is coordinated to six ligands with octahedral geometry. Four water molecules as well as the side chain oxygen atom of a serine residue from the P-loop and one oxygen atom from the β-phosphate bind to Mg^{2+} in the GDP structure. Two of the water molecules are replaced in the GTP structure by a threonine residue from switch I and an oxygen atom from the γ phosphate (similar to the arrangement shown in

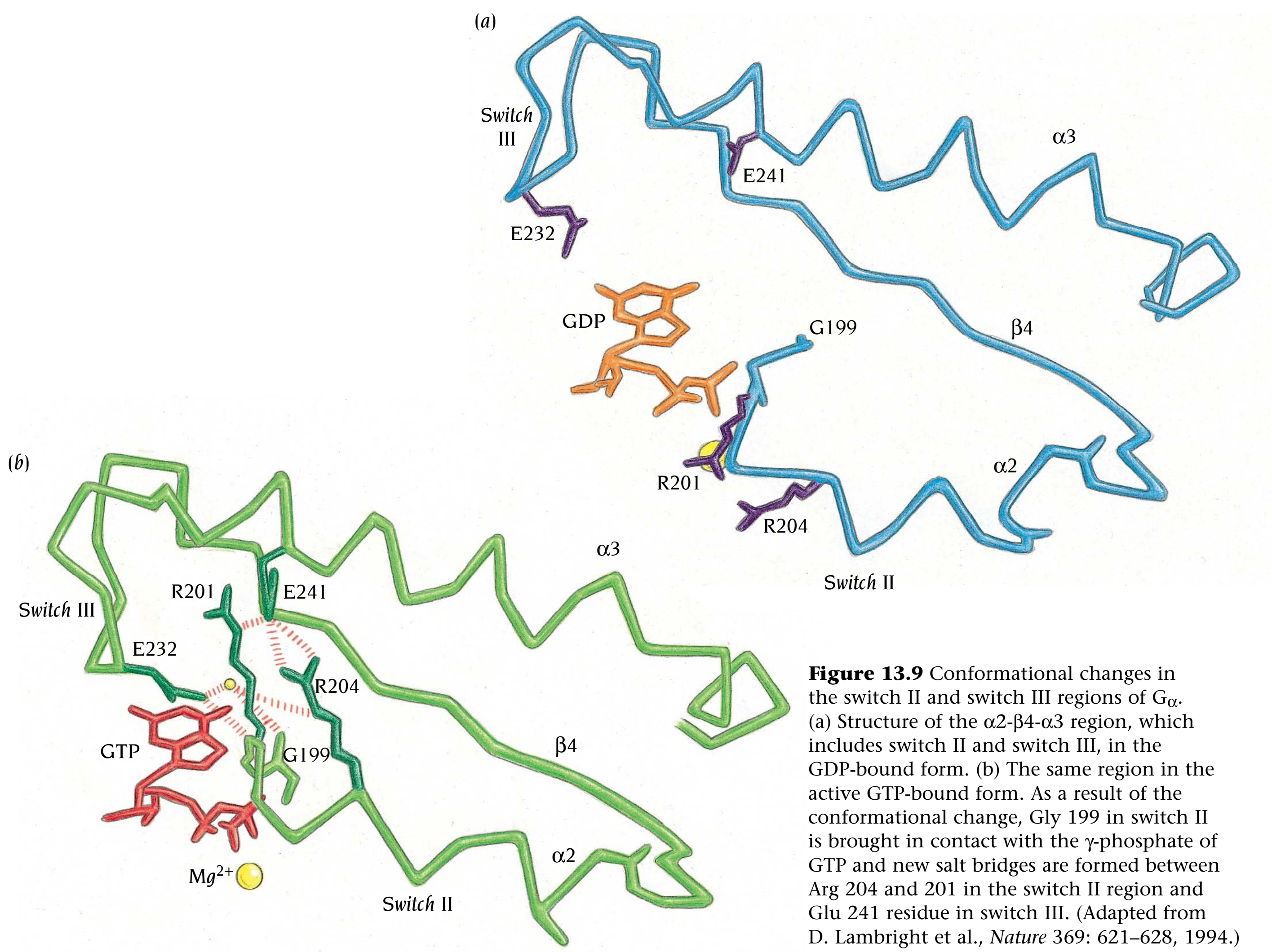


Figure 13.9 Conformational changes in the switch II and switch III regions of G_α. (a) Structure of the α2-β4-α3 region, which includes switch II and switch III, in the GDP-bound form. (b) The same region in the active GTP-bound form. As a result of the conformational change, Gly 199 in switch II is brought in contact with the γ-phosphate of GTP and new salt bridges are formed between Arg 204 and 201 in the switch II region and Glu 241 residue in switch III. (Adapted from D. Lambright et al., *Nature* 369: 621–628, 1994.)

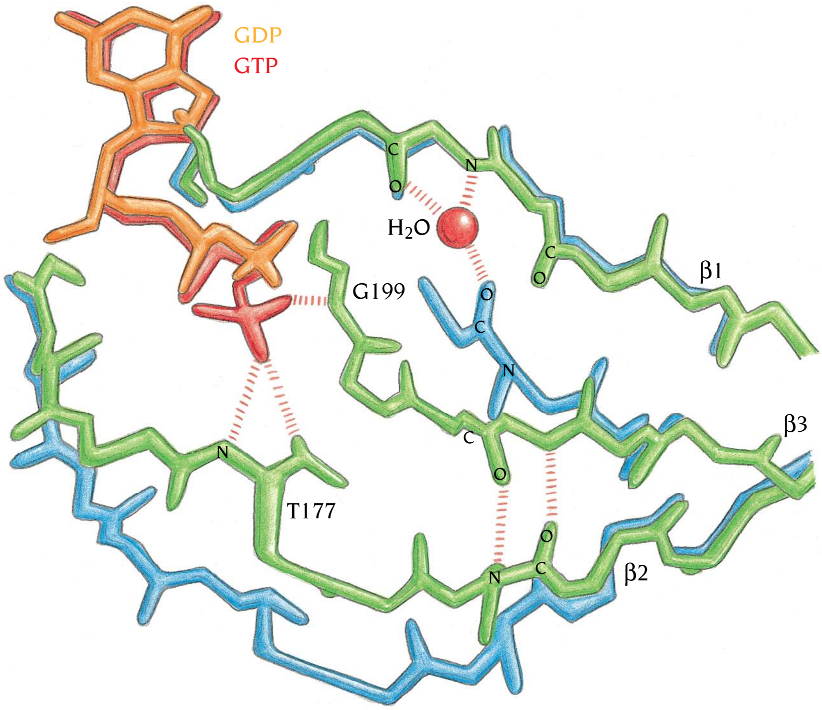


Figure 13.10 Rearrangements of the hydrogen bond network between strands 1, 2, and 3 in the β sheet of G_α as a consequence of the switch from the GDP (blue) to the GTP (green) conformation. Strand β3 pulls away from β1 and disrupts two hydrogen bonds in order to bring Gly 199 into contact with the γ-phosphate of GTP. As a consequence new hydrogen bonds are formed between β2 and β3. (Adapted from D. Lambright et al., *Nature* 369: 621–628, 1994.)

Figure 13.5). Structural changes in the switch I region are induced by hydrogen bonds to the γ phosphate from both the main chain N and side chain OH of the threonine residue, Thr 177, the oxygen of which is also coordinated to the Mg^{2+} ion (Figure 13.8). These structural changes are conceptually simple: after exchange of GDP for GTP, switch region I is drawn essentially as a whole towards the γ phosphate, bringing the side chain of Thr 177 into position for hydrogen bonding with the oxygen of the phosphate, and also into the vicinity of the Mg^{2+} where it replaces one of the water molecules observed in the GDP form (see Figure 13.8).

Changes in the switch II region are initiated by the formation of a hydrogen bond between the main chain NH of Gly 199 and the γ phosphate of GTP (Figures 13.8 and 13.9). Switch II comprises both the loop containing Gly 199 and helix α2. Movement in this loop is coupled to α2 so that to bring Gly 199 into this position, the region is stretched and rotated relative to the conformation in the GDP form (see Figure 13.7). As a consequence the end of β3 pulls away from β1, disrupting two hydrogen bonds (Figure 13.10). The new conformation is stabilized by the formation of additional hydrogen bonds between β3 and β2 and also by new salt bridges involving arginine residues in the switch II region and a glutamate residue in the switch III region (Figure 13.9).

The switch III region has no direct interactions with the γ phosphate, but forms a network of new highly conserved interactions with residues in switch II. The structural changes in switches I and II, which are triggered by the γ phosphate, have thus indirectly propagated to switch III. These three switches are clustered in one area of the G_α molecule that structural studies show is involved both in interactions with the β subunit of $G_{\beta\gamma}$ and with GTPase activator molecules. Structural changes in this region can therefore explain why the GTP form of G_α, but not the GDP form, binds to effectors and vice versa for binding to $G_{\beta\gamma}$.

GTPases hydrolyze GTP through nucleophilic attack by a water molecule

Some G proteins are slow GTP hydrolases with turnover numbers around two per minute, others such as Ras are only marginally catalytic. Kinetic experiments in solution have shown that in both cases the most likely mechanism of hydrolysis of GTP to GDP and phosphate is direct transfer of the γ phosphate to water without the formation of a covalently bound intermediate with the protein, as described for serine proteases in Chapter 11. This mechanism requires the GTPase to provide a catalytic base to remove a proton from a water molecule, producing a hydroxyl group that can directly attack the phophorous atom of the γ phosphate. GTP is not hydrolyzed in solution by negatively charged hydroxyl groups in the absence of the GTPase because the γ phosphate is negatively charged too. The GTPase must therefore also reduce the negative charge on the phosphate to catalyze hydrolysis. Ideally, to identify groups on the protein that carry out these two crucial functions we would determine the structure of the protein in complex with the transition state, but because transition states are very short-lived, even for slow enzymatic reactions, their structures cannot easily be studied. Instead, stable analogs that simulate the transition state are used; the complex between G_α, GDP and AlF_4^- is such an analog for the GTPase reaction (Figure 13.11).

The structures of two different G_α molecules complexed with AlF_4^- show only minor conformational changes compared with the structure of the ground state, represented by the complex with GTP-γS. These changes bring a glutamine side chain from the switch II region deeper into the active site closer to the γ phosphate. This glutamine residue acts as the catalytically required base when its oxygen atom accepts a proton from a water molecule, with the glutamine nitrogen atom donating a proton to the γ phosphate. This glutamine residue together with an arginine and a threonine residue from the switch I region reduce the negative charge of the γ phosphate through the formation of hydrogen bonds, thereby stabilizing the presumed transition state complex, as shown in Figure 13.12. The hydrogen atom taken from the water molecule to generate the attacking hydroxyl group is a part of this stabilizing network of hydrogen bonds. The phosphorous atom in the γ phosphate is assumed to be pentacoordinated, with the hydrogen bonds firmly stabilizing the geometry of the pentacoordinate γ phosphate.

All the residues involved in important functions in the catalytic mechanism are strictly conserved in all homologous GTPases with one notable exception. Ras does not have the arginine in the switch I region that stabilizes the transition state. The assumption that the lack of this catalytically important residue was one reason for the slow rate of GTP hydrolysis by Ras was confirmed when the group of Alfred Wittinghofer, Max-Planck Institute,

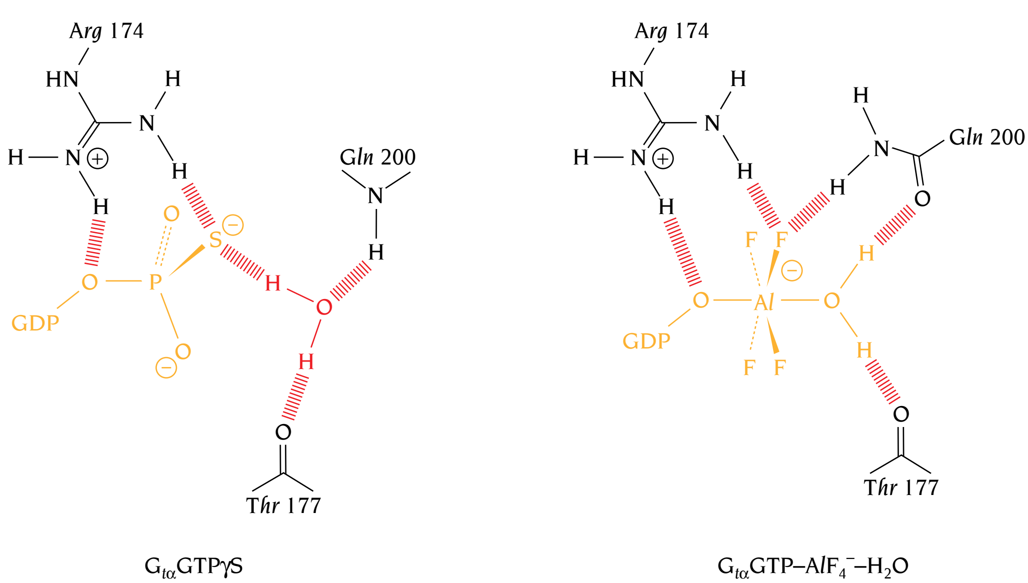


Figure 13.11 Interactions that are relevant for the GTPase mechanism, involving GTP-γS and GDP–AlF_4^-–H_2O at the active site of transducin G_α. GTP-γS mimics the ground state of bound GTP and GDP–AlF_4^-–H_2O mimics the transition state during catalysis. The positively charged guanidinium group of Arg 174 would increase the transfer of negative charge from the attacking hydroxyl to the β–γ bridging oxygen thereby facilitating hydrolysis of the terminal phosphate. (Adapted from J. Sondek et al., *Nature* 372: 276–279, 1994.)

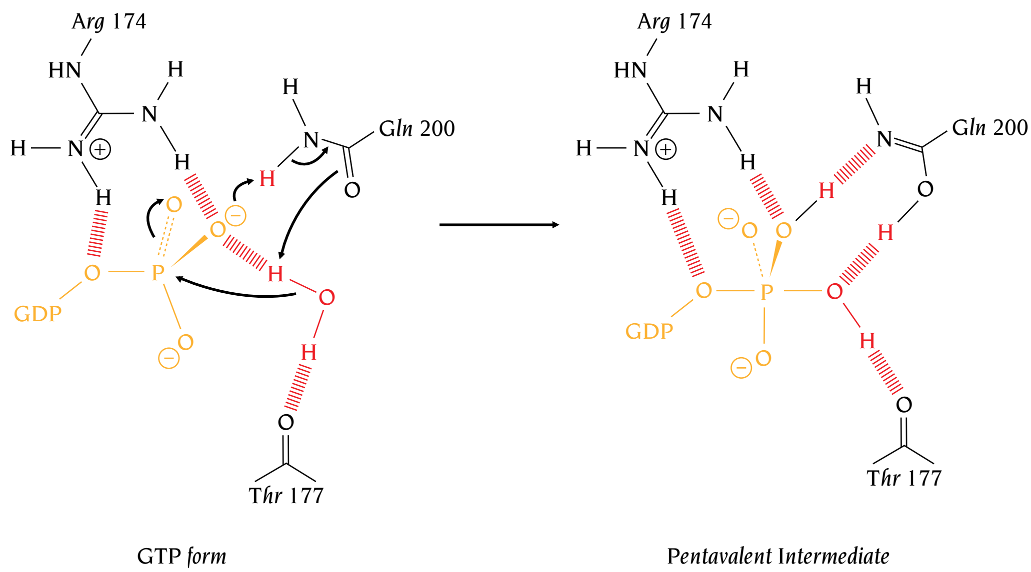


Figure 13.12 A concerted mechanism for GTP hydrolysis by G_α in which transfer of a proton to the γ phosphate is coupled to deprotonation of the attacking water by Gln 200. (Adapted from J. Sondek et al., *Nature* 372: 276–279, 1994.)

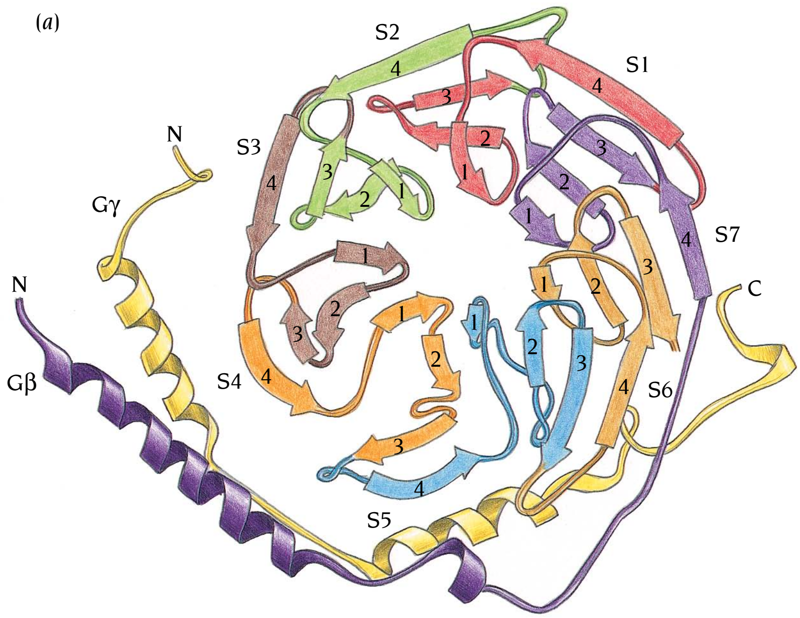
Dortmund, determined the structure of a complex between Ras, a transition state analog GDP. AlF_3 and the GTPase-activating protein GAP. GAP proteins accelerate the GTPase reaction of Ras nearly 100,000-fold. The structure of the Ras–GAP complex shows that GAP provides an arginine residue in the active site of Ras with the guanidinium group occupying a very similar position to that of the arginine in the switch I region in G_α. This arginine of GAP can increase the rate of the reaction by a factor of about 1000. The remainder of the rate increase results from the stabilization of the active conformations of switch regions I and II of Ras in the complex with GAP. The GAP binds to these regions such that the switches are locked in the structure that stabilizes the transition state of the reactants. The RGS molecules that regulate G protein signaling bind to G_α and enhance its rate of GTP hydrolysis by a similar mechanism. RGS molecules do not provide additional residues in the active site of G_α but rather bind to all three switch regions and stabilize their conformations.

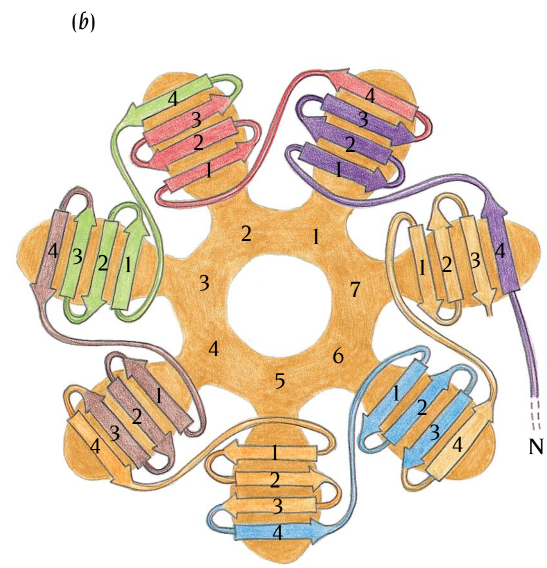
Disruption of the Ras–GAP relationship can have profound consequences for a cell, leading to unregulated cell division. For instance, about 25% of human tumors contain either of two mutated forms of Ras which resist GAP stimulation. In one of these oncogenic forms of Ras, the glutamine residue in the switch II region that stabilizes the transition state is mutated. In the other mutant the glycine residue in the P-loop, which is intimately packed against both the glutamine residue in Ras and the arginine residue from GAP, is changed. Any mutation of this residue changes the geometry of the GAP stabilized transition state, leading to loss of GAP-mediated activation of the catalytic activity of Ras.

In summary, structural studies of Ras and G_α with GTP-γS and a transition state analog have illuminated the catalytic mechanism of their GTPase activity, as well as the mechanism by which GTP hydrolysis is stimulated by GAP and RGS. In addition, these structural studies have shown how tumor-causing mutations affect the function of Ras and G_α.

The G_β subunit has a seven-blade propeller fold, built up from seven WD repeat units

The G_β polypeptide chain has about 350 amino acid residues whereas the γ chain has only about 70 residues. There are small variations in these lengths between species and between different gene products within the same species, but all known β chains and γ chains show significant sequence





homology among each other. Most of a β chain comprises seven repeats of a unit of approximately 40 amino acids. These repeated motifs are called WD repeats because they often end with a tryptophan, W, and an aspartate, D, residue. Such WD repeats occur not only in G proteins but also in effector molecules for signal transduction as well as in proteins involved in RNA processing, gene regulation, vesicular traffic, cell cycle regulation, and in a number of proteins of unknown function.

Structure determinations of $G_{\alpha\beta\gamma}$ by both Sigler's and Sprang's groups have shown that WD repeats form the blades of a β propeller structure, as described in Chapter 5 for neuraminidase, but with seven blades instead of the six β sheet blades of neuraminidase (Figure 13.13). Each blade consists of four antiparallel β strands forming a four-stranded β sheet. The seven blades are arranged like the blades of a propeller with the first β strand (β1) of each sheet close to the center of the structure and the last strand (β4) at the outer edge of the blade. Intuitively one would assume that each blade of four β strands would correspond to each WD sequence repeat. However, this is not the case. Instead, each WD sequence repeat forms the outermost β strand of one blade followed by the inner three strands of the following blade (Figure 13.13b). The outer strand of the seventh blade is provided by the N-terminal strand of the first WD repeat, thus linking the beginning and the end of the propeller. This arrangement of the β strands elegantly links the blades so that seven sequence repeat units form a circular superbarrel.

Analysis of the sequences of the seven repeats in G_β shows that there are five residues that are almost totally invariant (Table 13.2). These residues

Figure 13.13 (a) Schematic diagram of the $G_{\beta\gamma}$ heterodimer from transducin. The view is along the central tunnel. The seven four-stranded β sheets that form the seven blades of the propeller-like structure are labeled S1 to S7. The strands are colored to highlight the seven WD sequence repeats. The N-terminal α helices of the β and γ chains form a coiled coil. (b) Topological diagram of the seven-blade propeller domain of G_β. The strands are colored to highlight the seven WD sequence repeats. [(a) Adapted from J. Sondek et al., *Nature* 379: 369–374, 1996. (b) Adapted from D. Lambright et al., *Nature* 379: 311–319, 1996.]

Table 13.2 Sequence alignments of the seven WD repeats in the G_β subunit of transducin

Position numbers: 1, 5, 10, 20, 30, 40, 42; β strand arrows: 4, 1, 2, 3

Repeat	Sequence
WD-1	- - - - - - - - R T R R T L R G H L A K - - T Y A M H W G T D S R L L L S A S Q D G K L I I W D
WD-2	S Y T - - - - T N K V H A I P L R S S W - - V M T C A Y A P S G N Y V A C G G L D N I C S I Y N
WD-3	L K T R E G N V R V S R E L A G H T G Y - - L S C C R F L D D - N Q I V T S S G D T T C A L W D
WD-4	I E T - - - - G Q Q T T T F T G H T G D - - V M S L S L A P D T R L F V S G A C D A S A K L W D
WD-5	V R E - - - - G M C R Q T F T G H E S D - - I N A I C F F P N G N A F A T G S D D A T C R L F D
WD-6	L R A - - - - D Q E L M T Y S H D N I I C G I T S V S F S K S G R L L L A G Y D D F N C N V W D
WD-7	A L K - - - - A D R A G V L A G H D N R - - V S C L G V T D D G M A V A T G S W D S F L K T W N

Arrows denote β strands. Colored columns correspond to the five almost invariant residues described in the text and shown in Figure 13.14.

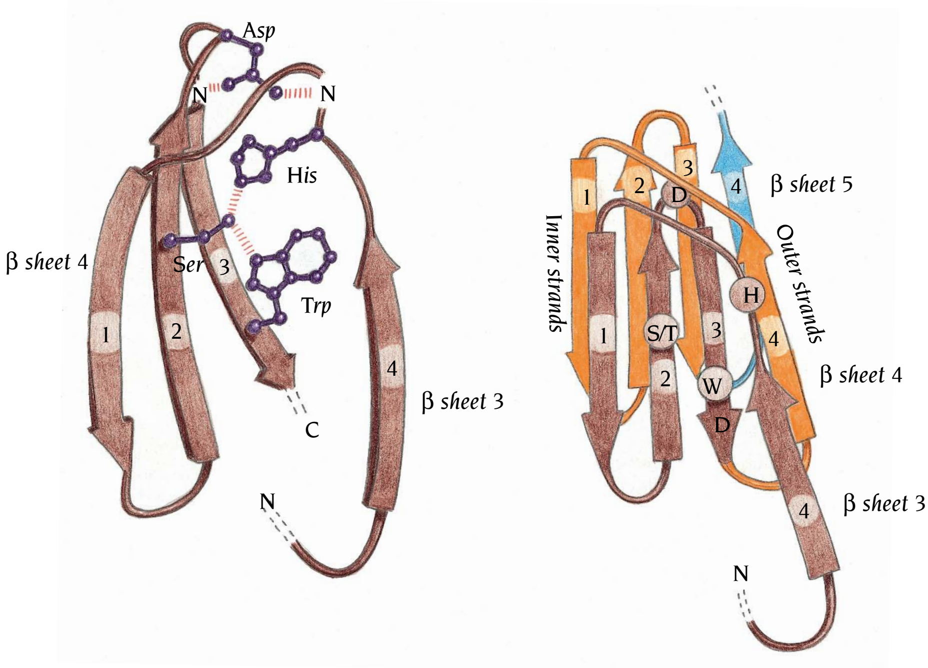


Figure 13.14 (a) Schematic diagram of the main chain and four almost invariant residues of the fourth WD repeat of G_β from transducin. The view is roughly perpendicular to the central tunnel and the plane of the sheet. The red stripes denote hydrogen bonds. (b) Schematic view of two WD repeats illustrating the structural relationships between two consecutive repeats. The first repeat is brown and the second repeat is orange. The positions of the four almost invariant residues in the first repeat are circled. (Adapted from J. Sondek et al., *Nature* 379: 369–374, 1996.)

occupy strategic positions to couple the outer strand of each blade to the inner three strands of the next blade (Figure 13.14). An Asp residue, which is present in all seven strands, is located in a tight turn between the middle two β strands. Side chain oxygens of this Asp form hydrogen bonds to main chain nitrogen atoms both in the tight turn and in the loop connecting the first and second strand of the same sequence repeat, which is the loop between two adjacent blades. This hydrogen bonding arrangement stabilizes and effectively couples the tight turn of one β sheet to the outer strand of the previous β sheet (Figure 13.14). A conserved His in the same loop between the blades participates in a hydrogen bond network to a Ser or Thr in the second strand which in turn forms a hydrogen bond to a Trp in the third strand. This spatial arrangement of conserved residues preserves a network of interactions that link the blades together in proper orientation. However, these conserved residues are not invariant, so there are small structural differences when one or the other conserved residue is absent.

The structure of $G_{\beta\gamma}$ (see Figure 13.13a) also explains why the dimer is stable whereas G_β alone is unstable and G_γ alone is unfolded *in vitro*. G_γ, which in the dimer is folded into two α helices in an extended arrangement lacking intrachain tertiary interactions (see Figure 13.13a), forms extensive, mainly hydrophobic interactions with G_β. The N-terminal α helix of G_γ forms a parallel coiled coil (see Chapter 3) with a long N-terminal α helix of G_β that precedes the seven WD repeats in the β-polypeptide chain. While these coiled-coil helices interact with one edge of the propeller region, the C-terminal half of G_γ is bound at the bottom side of the propeller structure as shown in Figure 13.13a. This unusual structure of G_γ is stabilized by the interactions with G_β; in the absence of these interactions the γ chain is unfolded and has no stable structure. The major side-chain interactions between G_β and G_γ involve residues that are conserved in different β and γ chains, so that the arrangement shown in Figure 13.13a is likely to be valid for all types of $G_{\beta\gamma}$ dimers.

The GTPase domain of G_α binds to G_β in the heterotrimeric $G_{\alpha\beta\gamma}$ complex

G_α-GDP binds to the $G_{\beta\gamma}$ dimer through its GTPase domain in a region of the β propeller opposite to where G_γ is bound (Figure 13.15). There are, therefore, no contacts between G_α and G_γ in the heterotrimeric $G_{\alpha\beta\gamma}$ complex. The

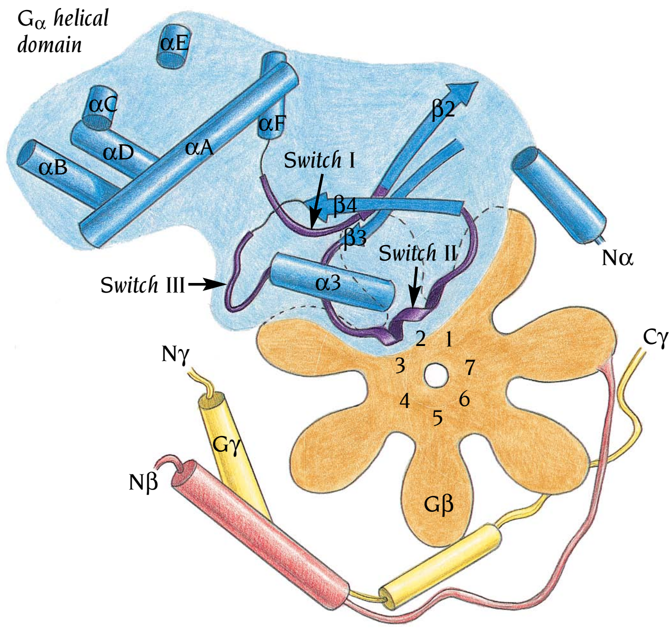


Figure 13.15 Schematic diagram of the heterotrimeric $G_{\alpha\beta\gamma}$ complex based on the crystal structure of the transducin molecule. The α subunit is blue with some of the α helices and β strands outlined. The switch regions of the catalytic domain of G_α are violet. The β subunit is light red and the seven WD repeats are represented as seven orange propeller blades. The γ subunit is yellow. The switch regions of G_α interact with the β subunit, thereby locking them into an inactive conformation that binds GDP but not GTP.

interactions between G_α and G_β occur at two distinct interfaces; these surfaces do not involve the helical domain of G_α, which is far from the β subunit and has no interactions with it. The more extensive interface involves residues in or adjacent to switch regions I and II of the GTPase domain of G_α, which interact with loops and turns on the top side of the β propeller. The second interface is formed between an N-terminal α helix of G_α and one edge of the β propeller. The N-terminal region in free G_α is disordered but it becomes stabilized into an α helix by interactions with the β subunit. Most of the contacts in the two interfaces involve residues that are highly conserved in both G_α and G_β. Substitutions at these positions are conservative even in sequences from distantly related organisms. It is, therefore, likely that the arrangement of subunits seen in Sigler's and Sprang's structures can be generalized to other members of the heterotrimeric G protein family.

The structure of $G_{\beta\gamma}$ is not changed at all on formation of the complex with G_α. The overall structure of G_α is also preserved except for the ordering of the N-terminal region into an α helix and significant conformational changes within the switch I and II regions (violet in Figure 13.15). The largest local structural changes in the switch interface occur in the switch II region, and involve those residues that interact with G_β directly. Some of these are conserved residues that are important to the function of G_α, for example Gly 199, which triggers conformational changes in the switch II region through hydrogen bonding with the γ phosphate of GTP, and Gln 200, which stabilizes the transition state for GTP hydrolysis (see Figures 13.8 and 13.9).

Binding to $G_{\beta\gamma}$ locks the flexible switch regions I and II of G_α into a conformation that firmly binds GDP but is nonproductive for GTP binding and hydrolysis. The replacement of GDP with GTP causes local but dramatic conformational changes to switch regions I and II, as shown in the G_α–GTP-γS structure, which disrupt nearly all of the contacts between $G_{\beta\gamma}$ and G_α in the switch interface, thereby triggering release of G_α from $G_{\beta\gamma}$ (see Figures 13.10 and 13.11).

The $G_{\beta\gamma}$ dimer appears to function as a rigid unit with critical residues positioned to interact with G_α–GDP. Whereas G_α activation proceeds through nucleotide-dependent structural reorganization, activation of $G_{\beta\gamma}$ occurs solely as a function of its release from G_α. As we will see, the G_α subunit acts as a negative regulator of $G_{\beta\gamma}$ by masking sites on the surface of $G_{\beta\gamma}$ that interact with downstream effector molecules.

Phosducin regulates light adaptation in retinal rods

One of the best characterized heterotrimeric G protein coupled pathways is the visual transduction system in rod cells, where rhodopsin absorbs photons through its retinal chromophore in a similar way as bacteriorhodopsin (see Chapter 12). Instead of initiating proton transfers through the membrane, as occurs in bacteriorhodopsin, rhodopsin activates the G protein transducin by catalyzing nucleotide exchange, leading to an activation of cyclic GMP phosphodiesterase, which degrades cyclic GMP. In a dark-adapted rod cell the absorption of one photon is sufficient to activate one rhodopsin molecule, leading to the degradation of more than 100,000 cyclic GMP molecules, enough to cause a neural impulse. This level of amplification is too high for daytime vision and a G protein regulator molecule called **phosducin** plays an important role in long-term light adaptation of rod cells.

Phosducin is expressed at high levels in retinal rods, and reaches a concentration similar to that of the G protein transducin. In light-adapted rods, unphosphorylated phosducin binds the $G_{\beta\gamma}$ of transducin with high affinity, sequestering $G_{\beta\gamma}$ and translocating it from the membrane to the cytosol. This prevents reassociation with G_{α}, thereby reducing signal amplification by transducin. In dark-adapted rods, phosducin is phosphorylated at serine 73, which reduces the stability of the phosducin–$G_{\beta\gamma}$ complex, allowing G_{α} to reassociate with $G_{\beta\gamma}$ and increase signal amplification. The crystal structure of a phosducin–$G_{\beta\gamma}$ complex, determined by Sigler's group explains some of these biological effects.

Phosducin binding to $G_{\beta\gamma}$ blocks binding of G_{α}

The phosducin polypeptide chain, of some 240 amino acids, is folded into two domains (Figure 13.16). The N-terminal domain is mostly α-helical and appears to be quite flexible since only a weak electron density is obtained in the structure determination. The actual path of the polypeptide chain from the end of helix A_{α} to the beginning of helix B_{α} is tentative due to slight disorder. This region is close to serine 73 at the beginning of B_{α}, which also becomes disordered on phosphorylation.

The C-terminal domain of phosducin is a five-stranded mixed β sheet with α helices on both sides, similar to the thioredoxin fold of disulfide isomerase DsbA described in Chapter 6. Despite significant sequence homology to thioredoxin, the phosducin domain, unlike other members of this family,

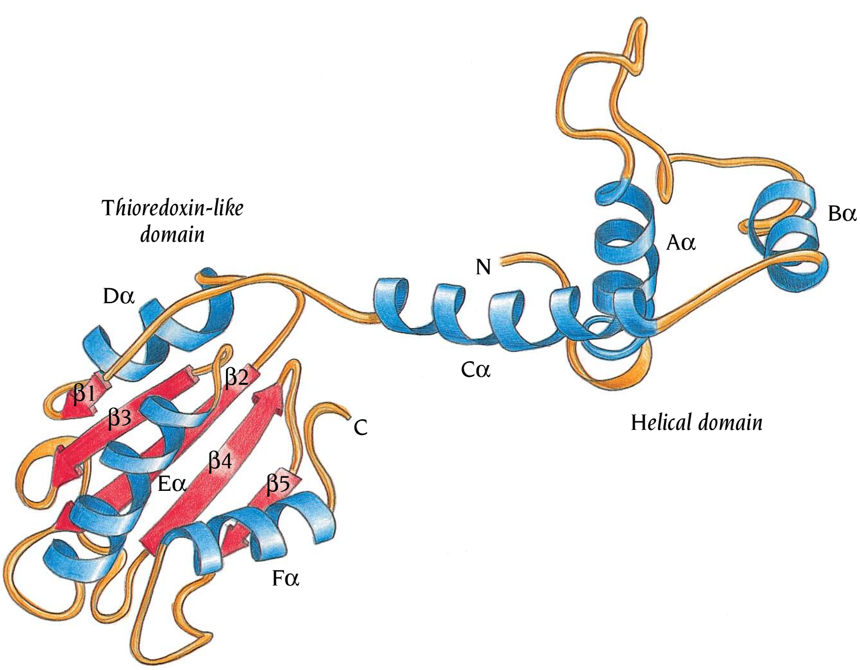


Figure 13.16 Schematic diagram of the phosducin molecule. Helices are blue, β strands are red and loop regions are orange. The structure folds into two separate domains, a N-terminal helical domain and a C-terminal domain that has the thioredoxin fold. Some of the loop regions in the helical domain are not well defined. (Adapted from R. Gaudet et al., *Cell* 87: 577–588, 1996.)

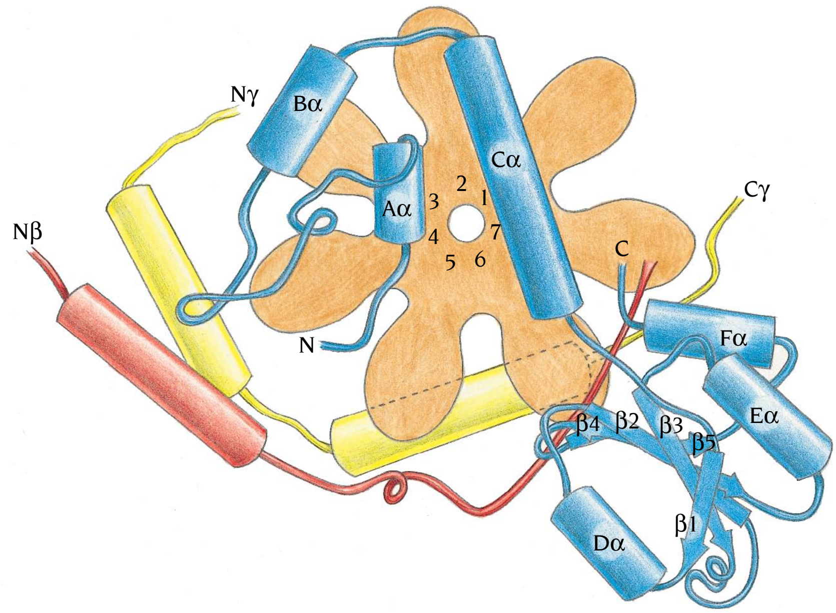


Figure 13.17 Schematic diagram of the structure of a complex between phosducin and the transducin $G_{\beta\gamma}$ dimer. The β subunit of transducin is light red and the seven WD repeats are represented as seven orange blades of a propeller. The γ subunit is yellow and the phosducin molecule is blue. The helical domain of phosducin interacts with G_β in the same region that G_α binds, thereby blocking the formation of a trimeric $G_{\alpha\beta\gamma}$ complex.

does not have a redox function because it contains only one of the two conserved cysteine residues that form the redox-active disulfide bridge.

In the complex with $G_{\beta\gamma}$ the two phosducin domains do not interact with each other, instead they wrap around the edge and the top side of the β propeller, to form an extensive interaction surface (Figure 13.17). The N-terminal domain of phosducin interacts with all of the top loops of the β propeller including part of the surface of $G_{\beta\gamma}$ that interacts with G_α (Figures 13.15 and 13.17). This interface between phosducin's N-terminal domain and $G_{\beta\gamma}$ clearly precludes association of the latter with G_α.

The C-terminal domain of phosducin binds to the edge of the β propeller opposite the β-γ coiled-coil region and close to the C-terminal region of the γ chain that carries the lipid anchoring $G_{\beta\gamma}$ to the membrane (see Figure 13.2). This surface of $G_{\beta\gamma}$ has also been implicated in receptor binding, raising the possibility that, by disrupting the orientation of $G_{\beta\gamma}$ relative to the membrane and the receptor, phosducin can interfere with the assembly and activation of the $G_{\alpha\beta\gamma}$ heterotrimer, in addition to blocking G_α binding to $G_{\beta\gamma}$.

In the structure of unphosphorylated phosducin that binds to $G_{\beta\gamma}$, Ser 73 points towards the flexible loop of phosducin and not towards $G_{\beta\gamma}$; it is, therefore, accessible on the surface for phosphorylation. Phosphorylation of Ser 73 cannot lead to the direct disruption of the phosducin/$G_{\beta\gamma}$ interaction. Rather, the structure suggests that phosphorylation may lead to conformational changes in the N-terminal domain of phosducin, especially in the flexible loop region, that could weaken or alter the phosducin/$G_{\beta\gamma}$ interface.

Two types of regulators of heterotrimeric G proteins have been identified. The RGS (regulators of G protein signaling) proteins act on the G_α subunits like GAP acts on Ras proteins, increasing their rate of GTP hydrolysis and thereby decreasing signal amplification as discussed above. By contrast, phosducin downregulates G protein function by acting on $G_{\beta\gamma}$ subunits, binding to $G_{\beta\gamma}$ and translocating it from the membrane to the cytosol, thereby preventing $G_{\alpha\beta\gamma}$ reassociation and subsequent rounds of receptor-mediated G_α activation. $G_{\beta\gamma}$ is an active component of the signal transduction system, interacting with numerous effector molecules so phosducin binding also directly inhibits the signaling pathway. As phosducin is present in various tissues and binds $G_{\beta\gamma}$ through conserved G_β residues, it is likely to be an important modulator of $G_{\beta\gamma}$ in several signal transduction pathways.

The human growth hormone induces dimerization of its cognate receptor

Peptide hormone receptors are sensory machines that direct cells to proliferate, differentiate, or even die. The growth hormone receptor belongs to one class of the cytokine or hematopoietic superfamily of hormone receptors that mediate signals from more than 20 known cytokines such as interleukin, erythropoietin and prolactin, as well as growth factors. Members of this receptor superfamily have a three-domain organization comprising an extracellular ligand-binding domain, a single transmembrane segment, and an intracellular domain that is quite diverse in sequence among the family members. The intracellular domain is structurally not well characterized, but it is known to associate reversibly with different cytoplasmic tyrosine kinases, which transmit the hormone-induced signal to activate a specific family of transcription factors. We shall describe here the crystal structure of the complex between the **human growth hormone**, **GH**, and the extracellular domain of its specific receptor, **GHR**, and compare this structure with that of growth hormone complexed with the **prolactin receptor**, **PLR**. These structures were determined by the group of Anthony Kossiakoff, Genentech, San Francisco.

The complex between GH and GHR contains one molecule of the hormone and two molecules of the receptor, even though the hormone does not have a pseudosymmetric structure with two similar binding sites. Instead, there are two completely different binding sites on the hormone, each of which binds to similar sites on the receptor molecules. These interactions are so far unique.

Like other hormones in this class of cytokines, GH has a four-helix bundle structure as described in Chapter 3 (see Figures 3.7 and 13.18). Two of the α helices, A and D, are long (around 30 residues) and the other two are about 10 residues shorter. Similar to other four-helix bundle structures, the internal core of the bundle is made up almost exclusively of hydrophobic residues. The topology of the bundle is up-up-down-down with two cross-over connections from one end of the bundle to the other, linking helix A with B and helix C with D (see Figure 13.18). Two short additional helices are in the first cross-over connection and a further one in the loop connecting helices C and D.

The two receptor extracellular domains in the complex are of course chemically identical and have very similar structures. Each polypeptide chain is arranged in two immunoglobulin-like domains joined by a linker region (Figure 13.19). Compared with an immunoglobulin constant domain (see Chapter 15) there is a "strand-switching" such that the edge of strand D, which forms hydrogen bonds to strand E in the immunoglobulin, is instead hydrogen bonded to strand C in the receptor and forms part of the sheet GFCD. This modified immunoglobulin fold is called a fibronectin III domain, since it also occurs in fibronectin.

The two domains are arranged with an acute angle between them so that loop regions from both domains come close together, forming the hormone-binding site (yellow in Figure 13.19). The loop regions are from different ends of the two domains; loops A–B, C–D and E–F in the N-terminal domain and loops B–C and F–G in the C-terminal domain. In contrast to the TIM-barrel domains (see Chapter 4), where the active site is always found at the same place in relation to the fold, immunoglobulin-like domains are much more promiscuous and utilize different areas of the fold for specific interactions with different molecules.

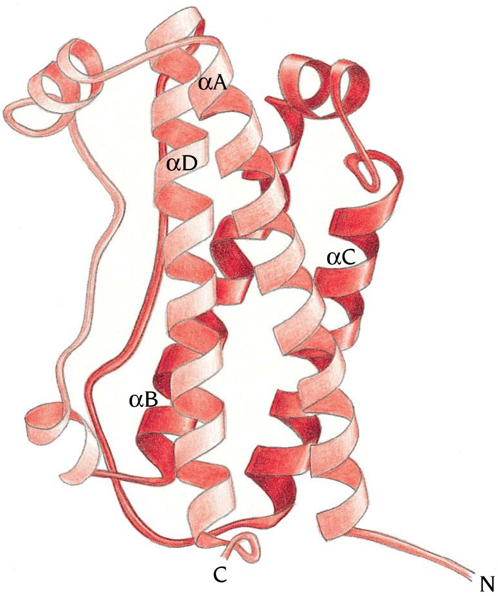


Figure 13.18 Ribbon diagram of the structure of human growth hormone. The fold is a four-helix bundle with up-up-down-down topology, and consequently there are two long cross-connections between helices A and B as well as between helices C and D. (Adapted from J. Wells et al., *Annu. Rev. Biochem.* 65: 609–634, 1996.)

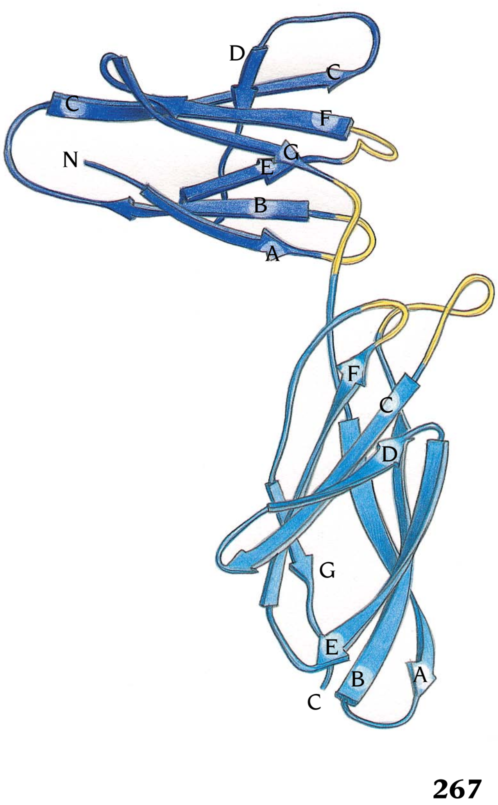


Figure 13.19 Ribbon diagram of the structure of the extracellular domain of the human growth hormone receptor. The hormone-binding region is formed by loops (yellow) at the hinge region between two fibronectin type III domains. (Adapted from J. Wells et al., *Annu. Rev. Biochem.* 65: 609–634, 1996.)

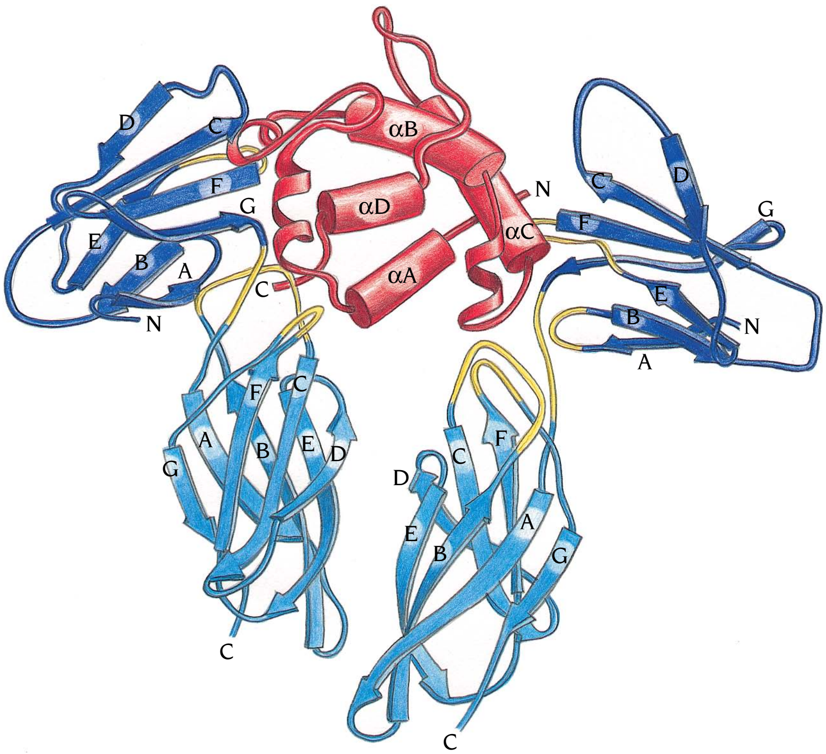


Figure 13.20 Ribbon diagram of the structure of a 1:2 complex between the human growth hormone and the extracellular domains of two receptor molecules. The two receptor molecules (blue) bind the hormone (red) with essentially the same loop regions (yellow). In contrast, the hormone molecule uses totally different surface regions to bind the two receptor molecules. (Adapted from Somers et al., *Nature* 372: 478–481, 1994.)

In the complex with the hormone, the two receptor domains form a dimer through direct interactions between their C-terminal regions and indirect interactions mediated by the hormone (see Figure 13.20). The two C-terminal domains are almost parallel, each having its C-terminus pointing away from the hormone towards the membrane. Biochemical and genetic studies have shown that dimerization of the receptor is the essential step responsible for activation of the intracellular domain. A reasonable speculation is that the close joining of the extracellular domains immediately above the membrane forces the transmembrane and intracellular domains into close proximity within and below the membrane, so as to initiate signaling.

Dimerization of the growth hormone receptor is a sequential process

The crystal structure of the GH–GHR complex shows that the receptor has a single ligand-binding site that can interact with either of two different sites on the hormone. Both receptor molecules in the complex use their A–B, C–D and E–F loops of the N-terminal domain as well as the linker and loops B–C and F–G of the C-terminal domain to form the binding site. Not only are essentially the same amino acid residues on both receptors used for binding, but the three-dimensional structures of these loop regions are also very similar. In contrast, the two binding sites on the hormone are very different; one site comprises residues from α helices A and D, and the other site is formed by residues from helices A and C (see Figure 13.20).

There is no structural similarity to these two binding sites; the A–C binding site is flat whereas the A–D site is concave, neither is there any similarity in the nature of the amino acids that form these two different binding sites. Nevertheless they bind to the same site on the receptor. However, the binding strengths of these two sites on the hormone are different. At high concentrations of the hormone a 1:1 complex is formed with receptor instead of the normal 1:2 complex. Experiments with mutants of the hormone show that this complex is formed by interactions to the A–D site on the hormone. Mutants in which binding to the A–C site is blocked still form the 1:1 complex whereas mutants that block binding to the A–D site do not form such complexes. These results nicely correlate to the structure of the complex. The A–D binding site involves 31 residues and the receptor covers an area of 1300 $Å^2$ of the hormone surface. The A–C site on the other hand only involves 24 residues and the binding area is 850 $Å^2$. Even though binding strength depends on the specific interactions involved there is in general a direct correlation between the size of the binding area and the strength of binding when proteins interact.

These results suggest that activation of the receptor by dimerization through binding the hormone is a sequential process. A 1:1 complex is first formed by hormone binding firmly to one receptor molecule through its A–D site. Once this complex is formed the binding of a second receptor to the A–C site of the hormone is facilitated by the interactions between the C-terminal domains of the two receptor molecules (see Figure 13.20). This latter interaction area is 500 $Å^2$, which together with the 850 $Å^2$ of the A–C site, provides a total binding area of 1350 $Å^2$, sufficient to stabilize the binding of the second receptor molecule to form the active complex.

The growth hormone also binds to the prolactin receptor

The prolactin receptor, PLR, which regulates milk production in mammals, belongs to the same receptor class as the growth hormone receptor. In addition to binding the hormone prolactin, PLR also binds and is activated by growth hormone. The extracellular domain of PLR forms a very stable 1:1 complex with growth hormone in solution: this complex has been crystallized and its structure determined (Figure 13.21). We shall compare this structure with the 1:2 complex of the same hormone with GHR.

The structure of the growth hormone in these two complexes is essentially the same, differing only in some of the loop regions and cross-over connections. The hormone interacts with the prolactin receptor through the same area that forms the strong A–D binding site to the growth hormone receptor. The structures of the individual domains of the prolactin receptor are very similar to the corresponding domains of the growth hormone receptor (even though their amino acid sequences share only 28% identity). The orientation of the domains with respect to each other is, however, significantly different. In spite of these differences the same loop regions of both

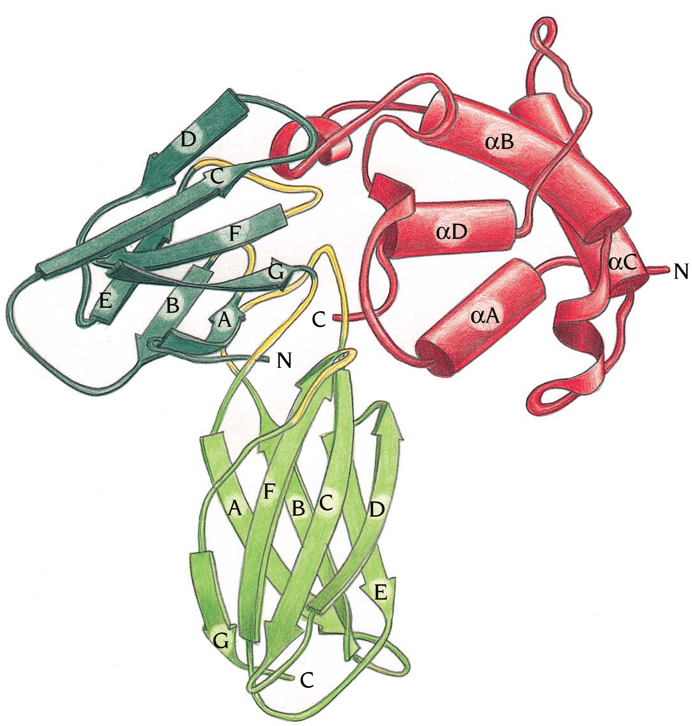


Figure 13.21 Ribbon diagram of the structure of a 1:1 complex between the human growth hormone (red) and the extracellular domain of the prolactin receptor (green). The hormone binds to the prolactin receptor using the same binding area as in the strong binding site to the growth hormone receptor. The two domains of the prolactin receptor have similar structures to the corresponding domains of the growth hormone receptor, but the orientation between the domains is different, causing differences in the details of the binding area. (Adapted from W. Somers et al., *Nature* 372: 478–481, 1994.)

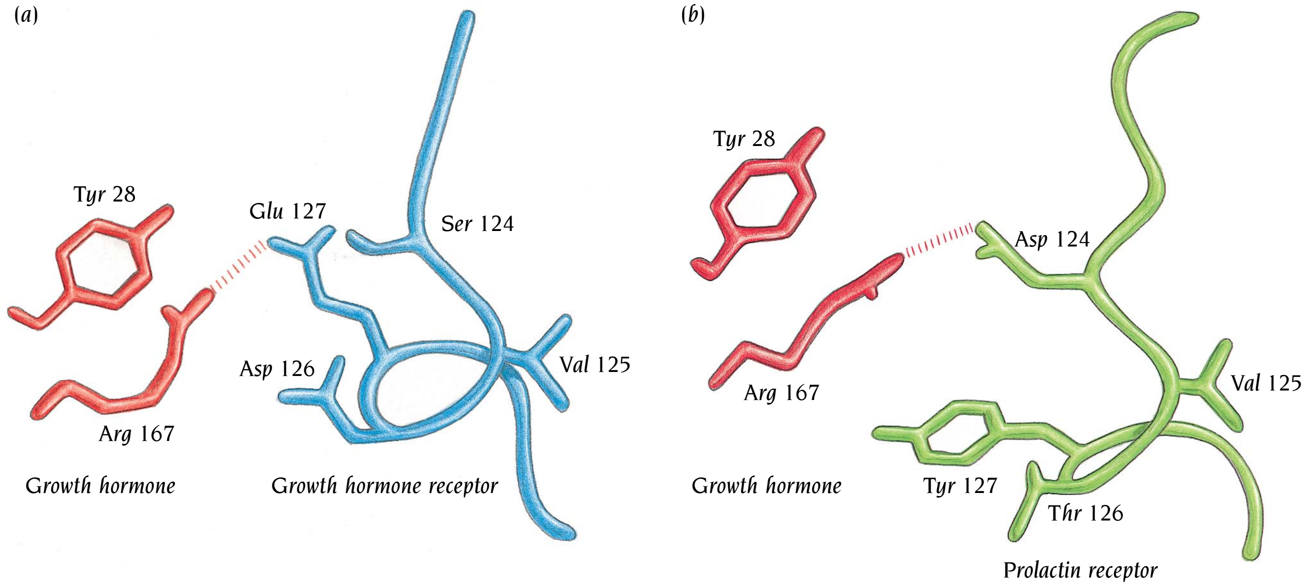


Figure 13.22 Hormone–receptor interactions involving the domain–domain linker region in the receptor. (a) Interactions between the growth hormone (red) and the growth hormone receptor (blue) linker region. Glu 127 of the receptor forms a salt bridge to Arg 167 in the hormone. (b) The same interaction area in the growth hormone (red)–prolactin receptor (green) complex. The displacement of the linker region due to differences in the domain orientations have brought Asp 124 in the prolactin receptor into contact with Arg 167 of the hormone. (Adapted from W. Somers et al., *Nature* 372: 478–481, 1994.)

receptors interact with the hormone. An example is given in Figure 13.22, which shows part of the interactions between growth hormone and the linker regions of the two receptors. The first four residues (124–127) of the linker regions are Ser-Val-Asp-Glu in GHR and Asp-Val-Thr-Tyr in PLR. The remaining seven residues of the linker regions are identical in the two molecules. The orientation of the linker in GHR is such that Glu 127 forms a salt linkage with Arg 167 of the hormone (Figure 13.22a). In PLR the corresponding Tyr 127 residue cannot form such a salt bridge. However, the different orientation of the linker region in the PLR complex removes the tyrosine from the interaction area and instead brings Asp 124 into a position where it forms a salt linkage with Arg 167 (Figure 13.22b).

A unique feature of the interaction of the hormone and PLR is at the beginning of the F-G loop in the C-terminal domain. In HGR the sequence is Arg-Asn-Ser whereas in PLR it is Asp-His-deletion. This loop interacts with His 18 and Glu 174 of the hormone. In PLR the orientation of this loop is such that the Asp and His residues, in combination with His and Glu from the hormone, form a strong binding site for a zinc atom that links the hormone and the receptor (Figure 13.23b). The presence of zinc increases the affinity of the hormone for the receptor *in vitro* by a factor of 10,000. As shown by mutagenesis studies His 18 and Glu 174 of the hormone are important for the tight binding to PRL but not to GHR.

These two examples demonstrate that both sequence differences and differences in the relative orientation of receptor domains are important factors for forming ligand-binding sites. This has considerable consequences for modeling binding sites of a receptor based on sequence homology with another receptor of known structure. Such modeling studies are frequently done, especially by drug companies, since many of these receptors are important targets for drugs. It is reasonably simple to model amino acid substitutions based on a structure with fixed main chain conformations. It is considerably more difficult to predict possible changes in the orientations of the domains due to sequence differences or ligand-induced structural changes.

Tyrosine kinase receptors are important enzyme-linked receptors

There are five known classes of **enzyme-linked receptors**: (1) receptor tyrosine kinases, which phosphorylate specific tyrosine residues on intracellular signaling proteins; (2) tyrosine kinase-associated receptors, such as the prolactin and growth hormone receptors we have already discussed, which

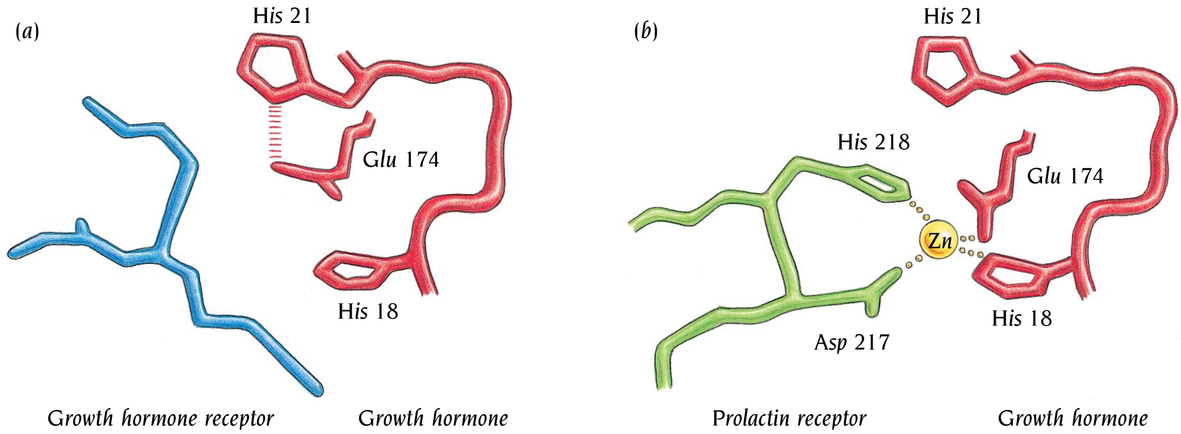


Figure 13.23 The F–G loop in the C-terminal domain of the prolactin receptor is involved in a unique interaction. (a) The F-G loop of the growth hormone receptor (blue) is not involved in any specific interactions with the growth hormone (red). (b) The F-G loop in the prolactin receptor forms a strong zinc-binding site that links the receptor (green) to the hormone (red). (Adapted from W. Somers et al., *Nature* 372: 478–481, 1994.)

associate with proteins that have tyrosine kinase activity; (3) receptor tyrosine phosphatases, which remove phosphate groups from tyrosine residues of specific intracellular signaling proteins; (4) transmembrane receptor serine/threonine kinases, which add a phosphate group to serine and threonine side chains on target proteins; and (5) transmembrane guanyl cyclases, which catalyze the production of cyclic GMP in the cytosol.

The protein kinase domains of groups 1, 2, and 4 all have homologous amino acid sequences and belong to one of the largest superfamilies known; the cyclin-dependent protein kinase CDK2 described in Chapter 6 (see Figure 6.16a) is one such member. In the complete yeast genome sequence more than 100 genes for members of this superfamily have been identified by sequence similarities, and it has been estimated that there are about 1000 members in humans. The first two types of receptors are by far the most numerous and they are thought to work in a similar way: ligand binding induces the receptors to oligomerize, which activates the tyrosine kinase activity of either the receptor itself or its associated nonreceptor tyrosine kinase. When activated, receptor tyrosine kinases usually cross-phosphorylate themselves on multiple tyrosine residues, with the phosphotyrosine residues then serving as docking sites for intracellular proteins. This results in the formation of an ensemble of proteins immobilized at the cell membrane, bringing together a diverse range of proteins such as kinases, phosphatases, phospholipases and G proteins such as Ras. In this way, a multisignaling complex is activated from which the signal spreads from the membrane to the cell interior.

Signaling through tyrosine kinase domains is involved in a variety of diverse biological processes including cell growth, cell shape, cell cycle control, transcription, and apoptosis (programmed cell death). Receptors regulating cell growth and differentiation were among the first to be studied, and they show a similar overall structural organization (Figure 13.24). The cytosolic region has a tyrosine kinase domain of 250–300 amino acid residues and, in addition, regions that contain tyrosine residues, in different sequence environments, which bind to different adaptor proteins when phosphorylated. The extracellular domains are different for different subclasses of these receptors, but are in many cases built up from immunoglobulin or fibronectin domains. A single transmembrane α helix links the two domains.

The second group, the tyrosine kinase associated receptors, have cytosolic domains that lack a defined catalytic function. This large and heterogeneous group includes receptors for cytokines as well as for some hormones including growth hormone and prolactin, discussed earlier. These receptors work through associated cytosolic tyrosine kinases, which phosphorylate various target proteins when the receptor binds its ligand. The best characterized of these associated tyrosine kinases are members of the **Src** family, so called because the first identified member of this family, the transforming agent of Rous sarcoma virus, induces sarcoma tumors of connective tissues. The structure and regulation of Src will be described later in this chapter.

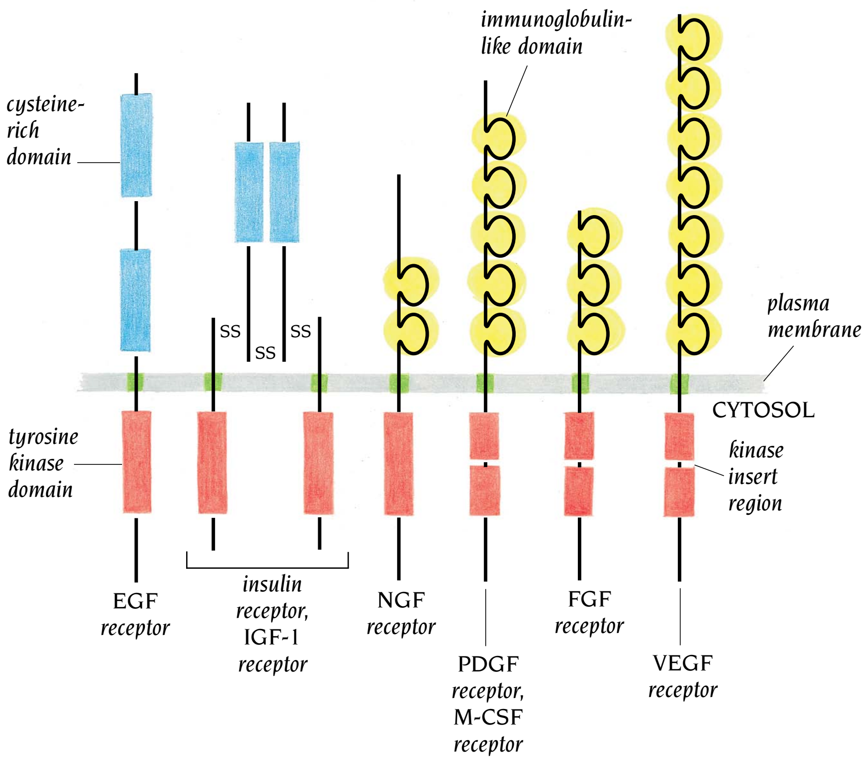


Figure 13.24 Six subfamilies of receptor tyrosine kinases involved in cell growth and differentiation. Only one or two members of each subfamily are indicated. Note that the tyrosine kinase domain is interrupted by a "kinase insert region" in some of the subfamilies. The functional significance of the cysteine-rich and immunoglobulin-like domains is unknown. (Adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)

Small protein modules form adaptors for a signaling network

How are the internal pathways controlled and organized in the complex network of signaling molecules involving more than a thousand different protein kinase domains and probably even more target molecules? A set of protein modules, which are covalently attached either to their associated protein kinases or to their target molecules, play an important role in regulating signal pathways (Figure 13.25a). These modules function as adaptors that bring together a kinase domain with its proper targets. Three important such modules are the **SH2** (Src-homology-2) and **SH3** (Src-homology-3) domains, whose names derive from the Src protein kinase molecule where they were first discovered, and the **PH** (pleckstrin-homology) domain, which was first defined as two repeated sequences in pleckstrin, the major substrate for Ser/Thr phosphorylation by protein kinase C in platelets. The SH2 and SH3 domains recognize short peptide motifs containing phosphotyrosine (pTyr)

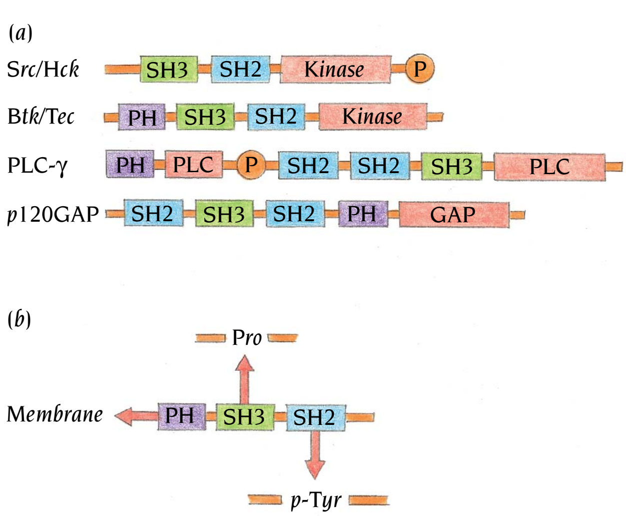


Figure 13.25 (a) Receptor-associated tyrosine kinases frequently contain within their polypeptide chains small modules such as Src homology domains, SH2 or SH3 and pleckstrin homology domains, PH. These modules function as adaptors to bring the kinase to its correct target molecule. (b) SH2 domains bind phosphotyrosine-containing peptide regions of the target molecules, SH3 domains bind proline-rich peptide regions of the target molecules and PH domains are involved in membrane association of kinases with their target molecules.

in the case of SH2, and proline-rich motifs in the case of SH3 (Figure 13.25b). The PH domains appear to mediate membrane association of kinases with their target molecules, in some cases by interaction with the headgroup of a phosphoinositide lipid.

SH2 domains bind to phosphotyrosine-containing regions of target molecules

The structures of many SH2 domains have been determined both by x-ray crystallography and by NMR. The overall fold of these domains is very similar, but residues involved in peptide binding differ, giving recognition specificity to SH2 domains from different signaling molecules. Figure 13.26 shows the crystal structure of the Src SH2 domain complexed with a peptide, determined by the group of John Kuriyan at the Rockefeller University, New York. The basic structure of the SH2 domain is an essentially antiparallel β sheet flanked by two helices. The β sheet comprises three long antiparallel β strands βB–βD and in addition two short strands βA and βG, which are hydrogen bonded in a parallel fashion to βB. These secondary structure elements are connected by relatively short loops, except the loop between βD and αB, which includes a small antiparallel β sheet. Hydrophobic cores are formed by packing the two α helices against each side of the β sheet.

The peptide binds to a relatively flat surface created by the two helices and the edge of the β sheet. Screening different phosphotyrosine-containing peptides showed the motif p-Tyr-Glu-Glu-Ile to be optimal for binding to Src. The peptide that Kuriyan studied contains this motif, and the structure shows that the phosphotyrosine moiety is firmly anchored to SH2 by three positively charged residues (green in Figure 13.26): two arginines from αA and βB and a lysine from βD. The two Glu residues of the peptide make weak

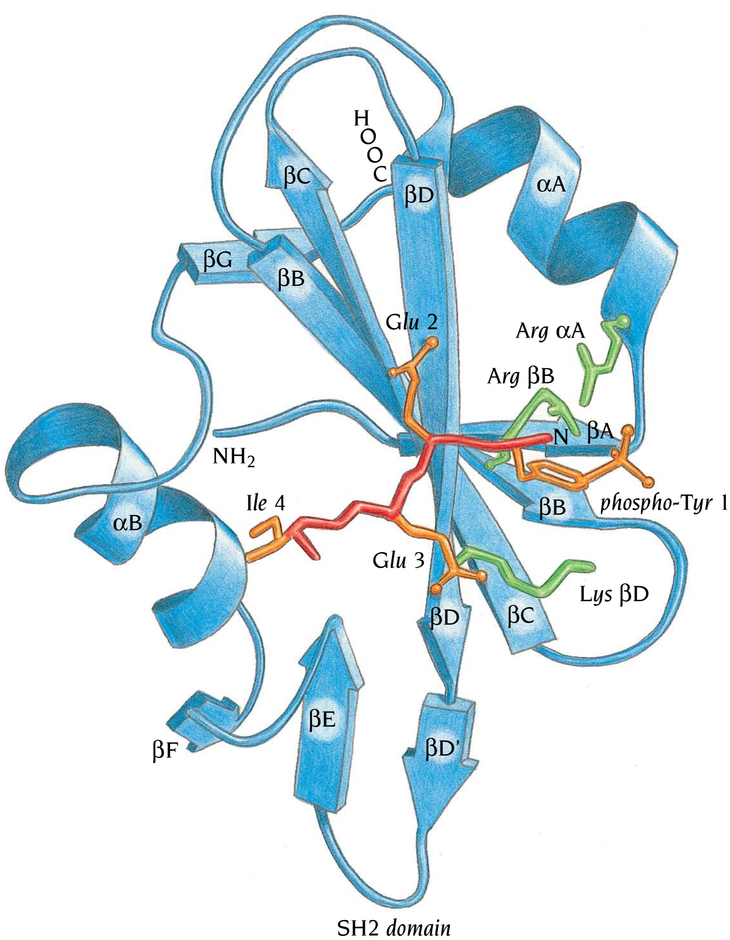


Figure 13.26 Schematic diagram of the SH2 domain from the Src tyrosine kinase with bound peptide. The SH2 domain (blue) comprises a central β sheet surrounded by two α helices. Three positively charged residues (green) are involved in binding the phosphotyrosine moiety of the bound peptide (red). (Adapted from G. Waksman et al., *Cell* 72: 779–790, 1993.)

(a) (b)

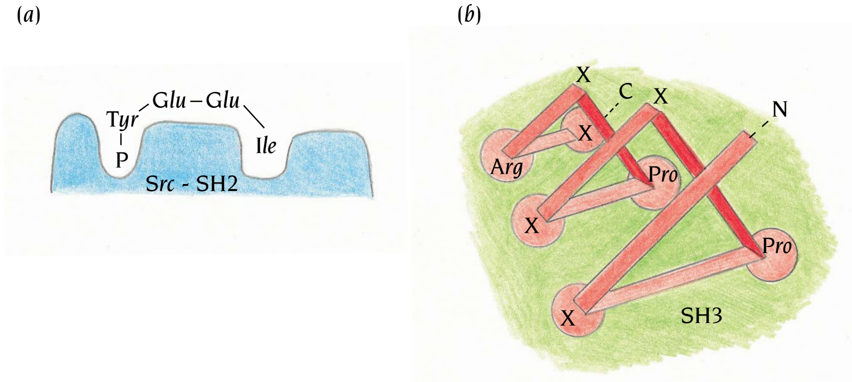


Figure 13.27 (a) All SH2 domains have a binding pocket with strong affinity for phosphotyrosine. Src-SH2 binds strongly to peptides containing two polar side chains followed by an aliphatic residue, after the phosphotyrosine. (b) SH3 domains bind proline-rich peptide regions with the consensus sequence X-Pro-X-X-Pro-X-Arg where X is any residue. These peptide regions bind in a helical conformation called polyproline II, which has three residues per turn. Such a helix has three ridges of side chains, two of which contact the binding surface of SH3 (green).

yet specific interactions with polar side chains of SH2, but the Ile residue is bound tightly to SH2 inside a hydrophobic pocket. This so-called pY+3 pocket is critical for determining peptide specificity and is formed by side chains from αB and βD, with loop regions from αB–βG and the small sheet βE–βF forming "jaws" that residue pY+3 fits between. The general hydrophobic character of the pocket is conserved in all SH2 domains, but the precise nature of the specific side chains that form the pocket in different SH2 domains determines the specificity of binding to target molecules.

The Src SH2 domain typifies a large number of those characterized to date. The pTyr fits into a pocket on the opposite side of the central sheet to the pY+3 pocket (Figure 13.27a). All known SH2 domains bind pTyr in essentially the same way, but some have a different pattern of contacts for the residues that follow. For example, in the Grb2 SH2 domain, a tryptophan side chain from the small sheet fills the pY+3 pocket, and the bound peptide takes a different course, with important interactions to an asparagine at pY+2. Screens of peptide libraries have detected the importance of this asparagine. The SH2 domain from PLC-γ1 contacts five mainly hydrophobic residues that follow pTyr.

SH3 domains bind to proline-rich regions of target molecules

The structures of many SH3 domains have also been determined; as an example, Figure 13.28a shows the SH3 domain from the tyrosine kinase Fyn, a member of the Src family, as determined by the group of Ian Campbell, Oxford University, using NMR methods. The SH3 domain consists of a five-stranded up-and-down antiparallel β structure, twisted into a barrel comprising two antiparallel β sheets that pack against each other so that their strands are nearly orthogonal. Strand β2 takes part in both sheets. This fold is common to all SH3 modules. The loop regions differ, however, and in some SH3 domains they contain small regions of secondary structure.

The peptide-binding site is a hydrophobic groove flanked by the RT loop between β1 and β2 and the n-Src loop between β3 and β4 (see Figure 13.28a). The latter is so named because neuronal Src has an insertion of six residues in this loop. The groove is lined with conserved aromatic residues.

SH3 binding elements in target molecules consist of proline-rich peptides. Upon binding, these segments adopt a left-handed polyproline type II helix with three residues per turn (see Figure 13.27b)—a structure discussed further in Chapter 14. The peptide ligand therefore has three ridges, two of which contact the SH3 domain (Figure 13.28b). Combinatorial screening studies (see Chapter 17) have shown that SH3 domains recognize sequences of about 10 residues. In the case of the Src SH3 domain, two classes of consensus motif were found: RXLPPLPXX (class I) and XXXPPLPXR (class II), where X is any residue. These were shown by subsequent NMR analysis to lie in opposite directions in the SH3 binding groove. Because proline is an imino acid, polyproline II helices with opposite N to C polarities present very similar surfaces. The ridges of these surfaces fit into notches formed by the aromatic

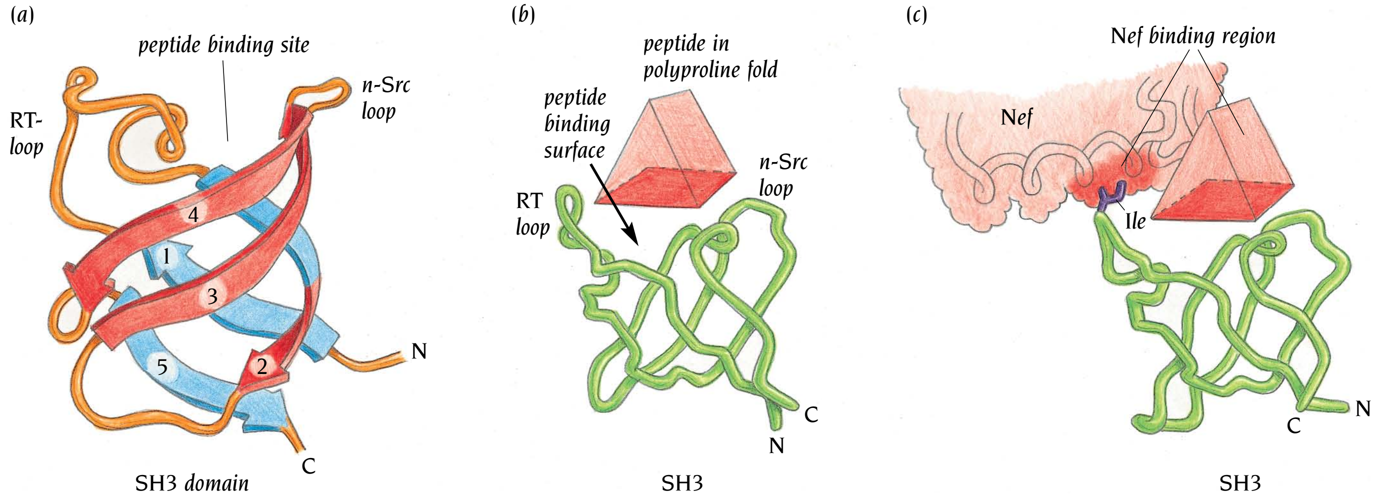


Figure 13.28 Schematic diagrams of the structure and binding modes of SH3 domains. (a) The polypeptide chain of the SH3 domain is folded into a five-stranded antiparallel β barrel with orthogonal packing of the β sheets. The peptide-binding site is at one end of the barrel, flanked by two loop regions, the RT-loop and the n-Src loop. (b) Isolated peptides (red triangular prism) bind rather weakly to SH3 even in the form of a polyproline helix. (c) Complete target molecules with the same peptide region bind much more strongly to the same SH3 module. The target molecule, Nef (red), which is involved in AIDS, interacts strongly with an Ile residue (violet) in the RT-loop of SH3, in addition to the proline-rich peptide binding (red triangular prism). [(a) Adapted from C.J. Morton et al., *Structure* 4: 705–714, 1996. (c) Adapted from W. Lim, *Structure* 4: 657–659, 1996.]

side chains in the SH3 groove. The polarity of binding is determined by additional interactions at one end of the groove. In the case of Src SH3, a negatively charged pocket helps to select the orientation of the bound peptide by receiving an arginine residue at one end or the other of the proline-rich segment.

The interactions of SH3 domains with their targets are strikingly promiscuous and relatively weak. Specificity is provided either by parallel contacts involving additional interacting modules (Grb2, for example, contains two SH3 domains, which bind to neighboring proline-rich sequences on its target), or by SH3–target interactions outside the polyproline helix, elegantly illustrated by the HIV protein, Nef. Nef is a regulatory protein encoded in the HIV genome; it is critical for viral pathogenesis and progression to AIDS. Nef has a high affinity for the SH3 domain of the Src family member Hck but not for the SH3 domain of the closely related kinase, Fyn. A key determinant of this specificity is an isoleucine residue, present in Hck but not in Fyn. Mutation of the corresponding Fyn residue (Arg 96) to Ile produces high specificity for Nef. John Kuriyan has determined the crystal structure of a complex between Nef and this mutant Fyn-SH3. As expected, a P-X-X-P element in Nef lies in the binding groove of the SH3 domain (Figure 13.29). The key Ile residue, which lies in the RT-loop of the SH3, forms a hydrophobic contact with side chains from two α helices in Nef (Figures 13.28b,c). Moreover, the polyproline helix itself is stabilized within the Nef molecule by packing against an α helix. Thus, the polyproline element is pre-formed, saving the free-energy "cost" of folding a disordered segment and increasing its affinity for the SH3 domain. Interactions of SH3 domains with their cellular targets are generally weak in order to be able to regulate the signal-transduction pathways in which they participate (which we discuss below). That is, the interaction must be readily reversed. Nef appears to have evolved to intervene strongly and irreversibly in a signaling pathway, but its physiological target is still not known. Identification of its partner might lead to a new target for anti-HIV drugs.

Src tyrosine kinases comprise SH2 and SH3 domains in addition to a tyrosine kinase

The polypeptide chain of **Src tyrosine kinase**, and related family members, comprises an N-terminal "unique" region, which directs membrane association and other as yet unknown functions, followed by a SH3 domain, a SH2 domain, and the two lobes of the protein kinase. Members of this family can be phosphorylated at two important tyrosine residues—one in the "activation loop" of the kinase domain (Tyr 419 in c-Src), the other in a short

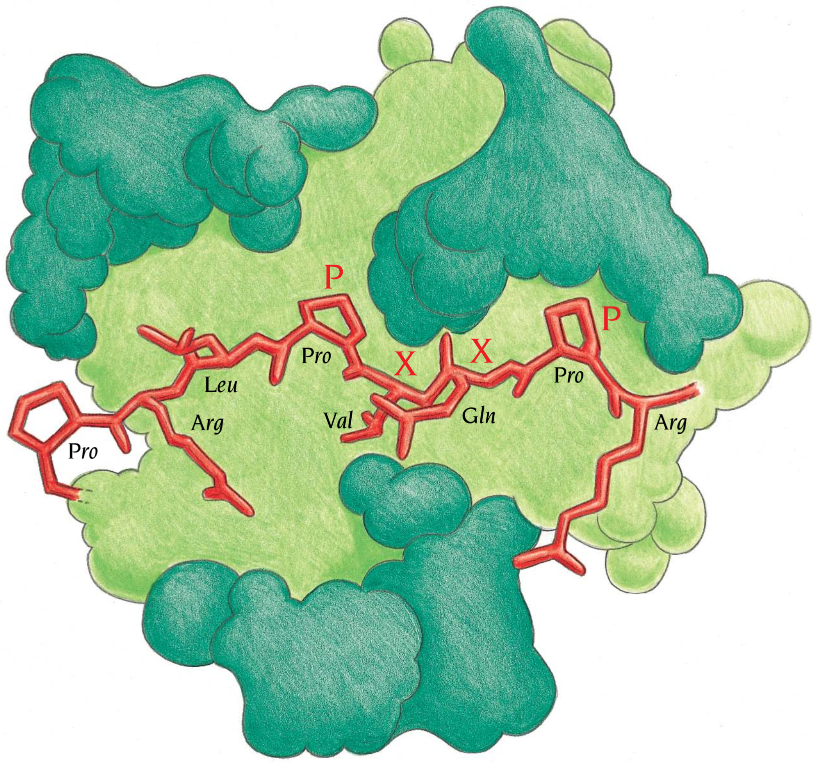


Figure 13.29 Schematic diagram of the proline-rich region of Nef bound to an SH3 domain. The peptide region (red) binds in a groove that extends across the surface of SH3 (green). (Adapted from W. Lim, *Structure* 4: 657–659, 1996.)

C-terminal tail (Tyr 527 in c-Src). Phosphorylation of Tyr 419 activates the kinase; phosphorylation of Tyr 527 inhibits it. Crystal structures of a fragment containing the last four domains of two members of this family were reported simultaneously in 1997—cellular Src by the group of Stephen Harrison and Hck by the group of John Kuriyan. The two structures are very similar, as expected since the 440 residue polypeptide chains have 60% sequence identity. The crucial C-proximal tyrosine that inhibits the activity of the kinases was phosphorylated in both cases; the activation loop was not.

The structures of each of the domains in the fragment (Figure 13.30) are very similar to previously described structures; the small N- and the large

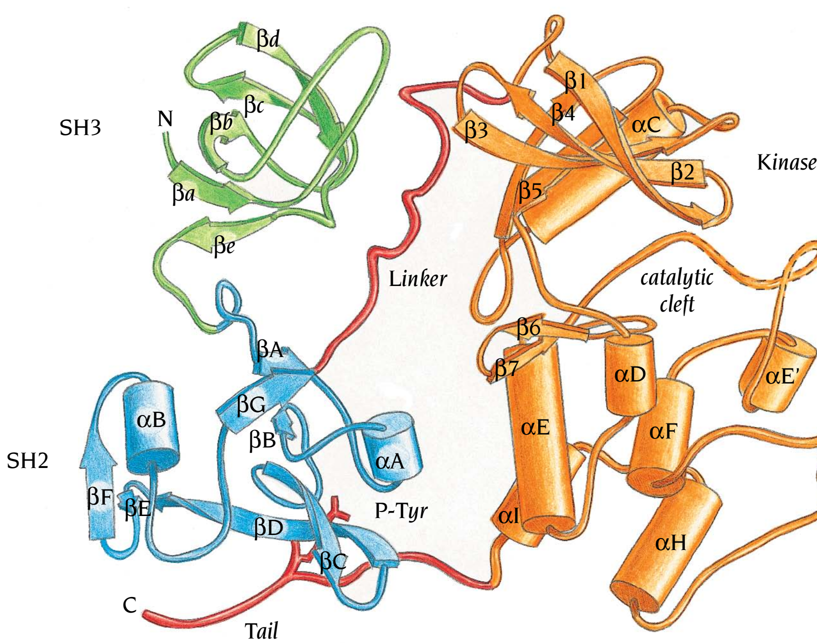


Figure 13.30 Ribbon diagram of the structure of Src tyrosine kinase. The structure is divided in three units starting from the N-terminus; an SH3 domain (green), an SH2 domain (blue), and a tyrosine kinase (orange) that is divided into two domains and has the same fold as the cyclin dependent kinase described in Chapter 6 (see Figure 6.16a). The linker region (red) between SH2 and the kinase is bound to SH3 in a polyproline helical conformation. A tyrosine residue in the carboxy tail of the kinase is phosphorylated and bound to SH2 in its phosphotyrosine-binding site. A disordered part of the activation segment in the kinase is dashed. (Adapted from W. Xu et al., *Nature* 385: 595–602, 1997.)

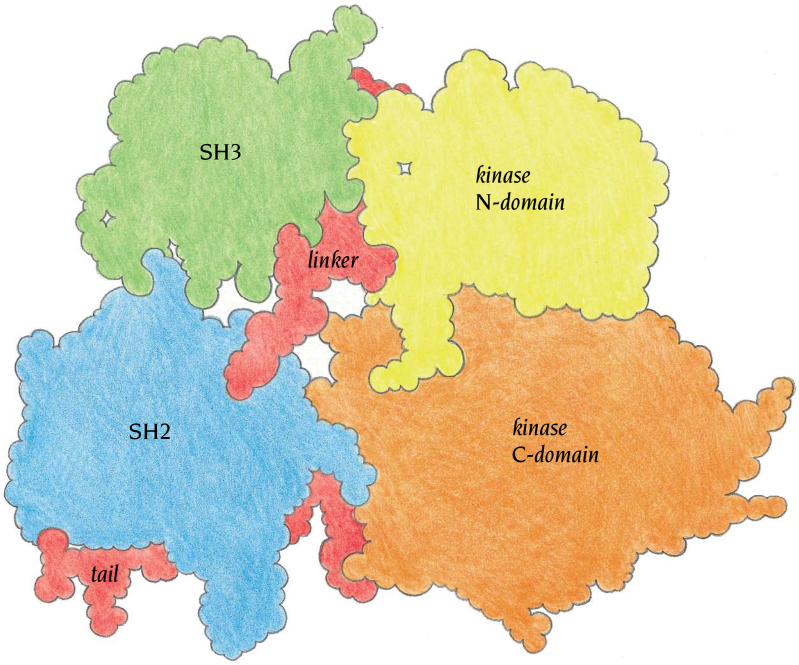


Figure 13.31 Space-filling diagram of Src tyrosine kinase in the same view as Figure 13.30. The SH2 domain makes only a few contacts with the rest of the molecule except for the tail region of the kinase. The SH3 domain contacts the N-domain of the kinase in addition to the linker region. There are extensive contacts between the N- and C-domains of the kinase. (Adapted from W. Xu et al., *Nature* 385: 596–602, 1997.)

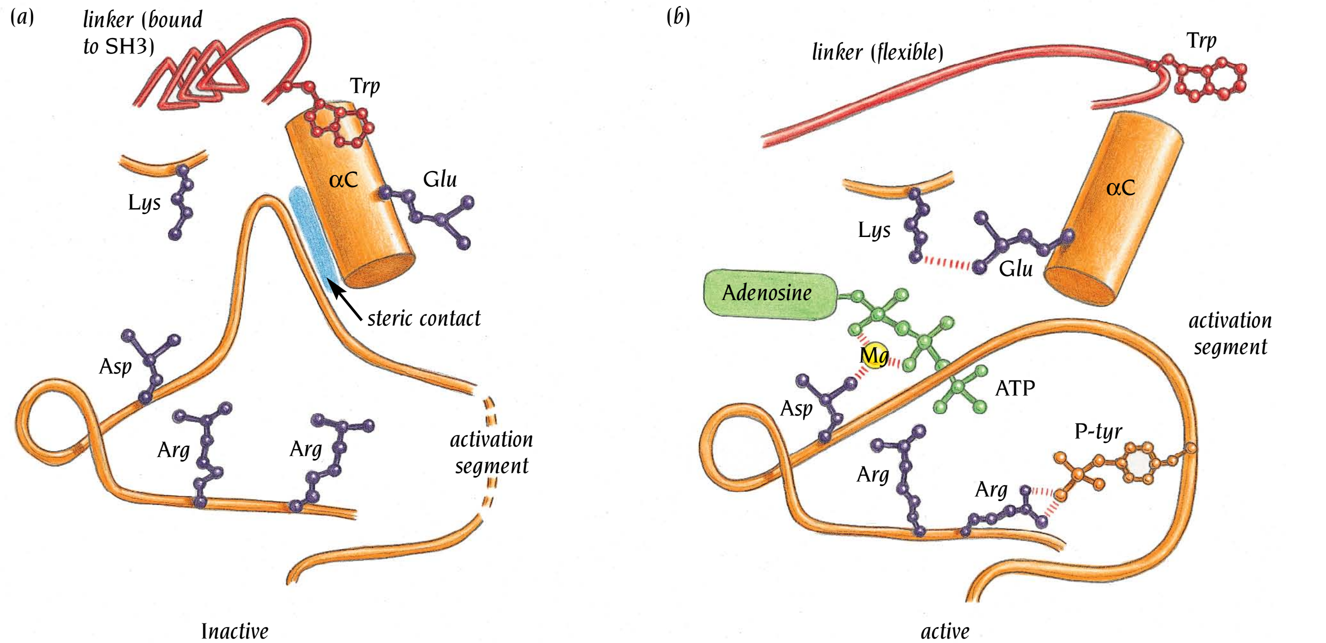
C-terminal lobes of the tyrosine kinase are similar to those of cyclin-dependent kinase described in Chapter 6 (see Figure 6.16a), while the SH2 and SH3 domains of Src and Hck have structures very similar to those of the isolated domains (see Figures 13.26 and 13.28a).

The inactive state of the kinase is a compact ensemble of the four domains. The SH2 and SH3 domains lie respectively beside the large and small domains of the tyrosine kinase, on the opposite side to the catalytic cleft. The phosphorylated C-terminal tail extends from the base of the large catalytic domain of the kinase into SH2, where the phosphotyrosine is bound in the usual SH2 binding site for phosphotyrosine. The mode of interaction between other residues of this tail and the binding groove of SH2 suggest a low-affinity interaction, because there is no side chain of the tail in the pY+3 pocket that binds Ile in high-affinity complexes of Src-SH2 (Figure 13.27a). The phosphorylated tail is a short but flexible tether that anchors the SH2 domain to the kinase. Apart from this tail, the SH2 domain makes only a few contacts with the catalytic domain of the kinase (Figure 13.31).

The linker region between the SH2 domain and the N-terminal domain of the kinase interacts with the groove of the SH3 domain. Residues 249–253 of the linker adopt a polyproline II helix structure in the SH3-binding groove even though the sequence around this region, -Pro-Gln-Thr-Gln-Gly-Leu-, deviates significantly from the consensus SH3 recognition sequence -P-X-X-P-X-R-. The proline residue occupies the position of the first proline in the consensus sequence, but the Gln residue cannot be accommodated in the binding pocket for the second proline, and so the course of the linker deviates from that of proline-rich peptides at this point. Contacts between the SH2 and SH3 domains are very limited (see Figure 13.31).

The two domains of the kinase in the inactive state are held in a closed conformation by assembly of the regulatory domains

A number of kinase structures have been determined in various catalytic states. For example, structures of the cyclin-dependent kinase, CDK2, in its inactive state and in a partially active state after cyclin binding have been discussed in Chapter 6. The most thoroughly studied kinase is the cyclic AMP-dependent protein kinase; the structure of both the inactive and the active



states have been determined by the group of Susan Taylor at the University of California, San Diego. In the inactive state the two domains of the kinase are wide apart, in an open conformation. In the active form the domains have moved toward each other and are in a closed conformation. A comparison of the phosphorylated Src kinase structure with these other kinases shows that the Src kinase domains are in a closed conformation but that their close approach is accompanied by a displacement and a rotation of the α helix, αC, in the N-terminal domain, away from its position in active kinases (Figure 13.32). Helix αC of Src is equivalent to the PSTAIRE helix in CDK2, where a similar rearrangement of the helix and the activation segment is seen in the inactive form (compare Figures 6.17 and 13.32). Because αC (like PSTAIRE) contributes to the active site of the enzyme, the observed, closed form of Src is catalytically inactive. What causes this inactivation, and how is the kinase activated?

Figure 13.32 Regulation of the catalytic activity of members of the Src family of tyrosine kinases. (a) The inactive form based on structure determinations. Helix αC is in a position and orientation where the catalytically important Glu residue is facing away from the active site. The activation segment has a conformation that through steric contacts blocks the catalytically competent positioning of helix αC. (b) A hypothetical active conformation based on comparisons with the active forms of other similar protein kinases. The linker region is released from SH3, and the activation segment changes its structure to allow helix αC to move and bring the Glu residue into the active site in contact with an important Lys residue. In the activation segment an Arg binds to phosphotyrosine, another Arg binds to the γ phosphate of ATP and an Asp binds to the Mg atom in the active site. (Adapted from F. Sicheri et al., *Nature* 385: 602–609, 1997.)

The interaction of the phosphotyrosine tail with SH2 is clearly important for inactivation of the tyrosine kinase, because regulation *in vivo* is lost upon dephosphorylation or mutation of Tyr 527. The observed interactions of the SH2 and SH3 domains with the linker, the phosphorylated C-terminal tail and the kinase itself, all serve to hold the kinase lobes together, thereby stabilizing ejection of the αC from the catalytic site (see Figure 13.32) as well as holding the activation loop in a conformation that protects Tyr 419 from phosphorylation. A key feature of the assembled regulatory apparatus is the position of the SH3 domain, which binds the linker regions between SH2 and the N-terminal lobe of the kinase and may affect the conformation of αC. Dephosphorylation of tyrosine 527 or recruitment of the SH2 or SH3 domains by strongly binding target molecules abolishes the intermolecular interaction of the linker region and presumably allows helix αC and the activation segment to shift into their active conformations. The SH2 and SH3 domains apparently not only function as adaptors between the kinase and target molecules but are also important for autoregulation of the intrinsic catalytic activity of Src-tyrosine kinases.

Conclusion

Signal-transducing receptors are plasma membrane proteins that bind specific molecules, such as growth factors, hormones, or neurotransmitters, and then transmit a signal to the cell's interior, causing the cell to respond in a specific manner. All membrane-bound receptor molecules have an extracellular domain that receives and recognizes a specific signal, a transmembrane region through which the signal is transmitted, and an intracellular domain that elicits a response. In many cases, the signal is amplified by a network of intracellular signaling pathways.

G proteins are molecular amplifiers for a large number of seven-transmembrane helix receptors that regulate responses like vision, smell and stress response. They are heterotrimeric molecules, $G_{\alpha\beta\gamma}$, that dissociate into membrane-bound G_α and $G_{\beta\gamma}$ signal transmitters upon activation of the receptor.

G_α has two domains: a helical domain and a catalytically active GTPase domain, which is similar to the small G protein, Ras. Both Ras and G_α switch between two conformations, an active GTP-bound form and an inactive GDP-bound form. Only the inactive GDP-bound form binds to $G_{\beta\gamma}$. Three switch regions have different conformations in the two forms. The cellular response of G_α and Ras depends on the length of time these molecules stay in the active GTP form—that is, on the catalytic rate of their GTPase activity. The rate of Ras GTPase activity is enhanced by binding to GAP, which provides a catalytically important arginine to the GTPase active site and stabilizes the switch regions in their active conformations. Mutations of Ras that abolish this rate enhancement are oncogenic because they render Ras constitutively active, promoting undifferentiated cell growth. The activity of G_α, which has a far higher catalytic rate than Ras due to the presence of an intrinsic corresponding Arg, is regulated by proteins that affect the conformation of the switch regions.

G_β has an N-terminal α helix that forms a coiled-coil structure with an N-terminal helix of G_γ in the $G_{\beta\gamma}$ dimer. The remaining part of G_β contains seven WD sequence repeats, which build up a circular seven-blade propeller structure, where each blade is an antiparallel β sheet of four strands. Each blade is formed by regions from two WD repeats, facilitating the formation of a circular structure. The GTPase domain of G_α binds to G_β in trimeric $G_{\alpha\beta\gamma}$ in such a way that the switch region of G_α is locked into the inactive form that binds GDP. The negative regulator of $G_{\beta\gamma}$ signaling, phosducin, binds to the same region of $G_{\beta\gamma}$ as G_α, thereby both preventing $G_{\beta\gamma}$ activating downstream signaling molecules and preventing reactivation of G_α by blocking the formation of the $G_{\alpha\beta\gamma}$ complex. Essentially all of the molecular mechanisms discussed (switch region conformational changes, hydrolysis, association and dissociation) are conserved throughout the trimeric G protein superfamily.

Growth hormones activate their cognate receptors by inducing the formation of receptor dimers. The structure of a complex between human growth hormone and the extracellular domains of the growth hormone receptor shows that one hormone molecule can bind two receptors. The hormone uses quite different surface areas for binding the same regions of the two receptor molecules. The structure of a complex between the growth hormone and the prolactin receptor has shown that subtle conformational differences can provide novel binding interactions, thereby increasing the range of possible interactions between a hormone and receptors.

Signaling through tyrosine kinase domains is involved in such diverse biological activities as cell growth, cell shape, cell cycle control, transcription and apoptosis. Receptor-associated cytosolic tyrosine kinases contain a set of protein modules such as SH2, SH3, and PH domains that function as adaptors to bring together a kinase domain and its appropriate target. SH2 and SH3 domains recognize short peptide sequences containing phosphotyrosine and proline-rich regions, respectively; some PH domains recognize lipid head groups. SH2 domains fold into a central β sheet surrounded by two helices and have a specific phosphotyrosine-binding pocket. SH3 domains fold into a β barrel structure with the peptide-binding site at one end of the barrel, wedged between two loop regions. The peptide region binds in a polyproline type II helix conformation with three residues per turn. Proline-rich sequences favor such a conformation. Additional specificity for target proteins can be provided by interactions between the loop regions and residues of the target protein outside the proline-rich sequence.

Src tyrosine kinase contains both an SH2 and an SH3 domain linked to a tyrosine kinase unit with a structure similar to other protein kinases. The phosphorylated form of the kinase is inactivated by binding of a phosphotyrosine in the C-terminal tail to its own SH2 domain. In addition the linker region between the SH2 domain and the kinase is bound in a polyproline II conformation to the SH3 domain. These interactions lock regions of the active site into a nonproductive conformation. Dephosphorylation or mutation of the C-terminal tyrosine abolishes this autoinactivation.

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Fibrous Proteins

14

Proteins are usually separated into two distinct functional classes: passive structural materials, which are built up from long fibers, and active components of cellular machinery in which the protein chains are arranged in small compact domains, as we have discussed in earlier chapters. In spite of their differences in structure and function, both these classes of proteins contain α helices and/or β sheets separated by regions of irregular structure. In most cases the fibrous proteins contain specific repetitive amino acid sequences that are necessary for their specific three-dimensional structure.

Fibrous proteins can serve as structural materials for the same reason that other polymers do; they are long-chain molecules. By cross-linking, interleaving and intertwining the proper combination of individual long-chain molecules, bulk properties are obtained that can serve many different functions. Fibrous proteins are usually divided in three different groups dependent on the secondary structure of the individual molecules: coiled-coil α helices present in keratin and myosin, the triple helix in collagen, and β sheets in amyloid fibers and silks.

The α-helical fibers of wool are flexible, can be extended up to twice their length, and are also elastic so that the fibers return to their original length when the tension is released. Collagen fibers by contrast are strong, resistant to stretching and relatively rigid. The β sheet fibers are both strong and very flexible. A prime example are the fibers of spiders' webs, some of which are stronger than steel wires of the same dimension, and at the same time are flexible enough not to break when an insect is caught in the web. Fibrous β sheets can also be formed *in vivo* by misfolding of soluble globular proteins, thereby causing conditions such as Alzheimer's disease and prion diseases.

Fibrous proteins often form protofilaments or protofibrils that assemble into structurally specific, higher-ordered filaments and fibrils. Examples include collagen, amyloids, intermediate filaments, tubulin, myosin, and fibrinogen. Such filaments or fibrils cannot be crystallized because they can only be ordered in two dimensions. However, ordered fibers give two-dimensional diffraction patterns from which information on their overall structure can be deduced (see Chapter 18). On the other hand, some fibrous proteins, or fragments thereof, such as fibrinogen, tropomyosin, collagens, spectrin, and myosin, have been crystallized and their detailed atomic structures determined. Such information has been useful for interpretation of diffraction patterns or electron micrographs of ordered filaments or fibers.

The assembly of fibrous proteins into higher-ordered structures is in some cases a dynamic process. Microtubules, which are long, hollow polymers of globular proteins, tubulins, continually depolymerize and repolymerize, thereby governing the location of organelles and other cell components. In fibroblasts the half-life of an individual microtubule is about 10 minutes, while the average lifetime of a tubulin molecule is more than 20 hours. Thus each tubulin molecule will be recycled through many microtubules during its lifetime.

Collagen is a superhelix formed by three parallel, very extended left-handed helices

The term **collagen** derives from the Greek word for glue and was defined in the 1893 edition of the Oxford Dictionary as "that constituent of connective tissue which yields gelatin on boiling." A modern definition, based on sequence determinations of different family members, would be that collagens are proteins that assemble into fibrous supramolecular aggregates in the extracellular space and which comprise three polypeptide chains with a large number of repeat sequences Gly-X-Y where X is often proline and Y is often hydroxyproline. Hydroxyproline is formed by the posttranslational modification of proline by a specific enzyme, prolyl hydroxylase, and is rarely found in other proteins. Each collagen polypeptide chain contains about 1000 amino acid residues and the entire triple chain molecule is about 3000 Å long.

Collagen chains are synthesized as longer precursors, called procollagens, with globular extensions—propeptides of about 200 residues—at both ends. These procollagen polypeptide chains are transported into the lumen of the rough endoplasmic reticulum where they undergo hydroxylation and other chemical modifications before they are assembled into triple chain molecules. The terminal propeptides are essential for proper formation of triple

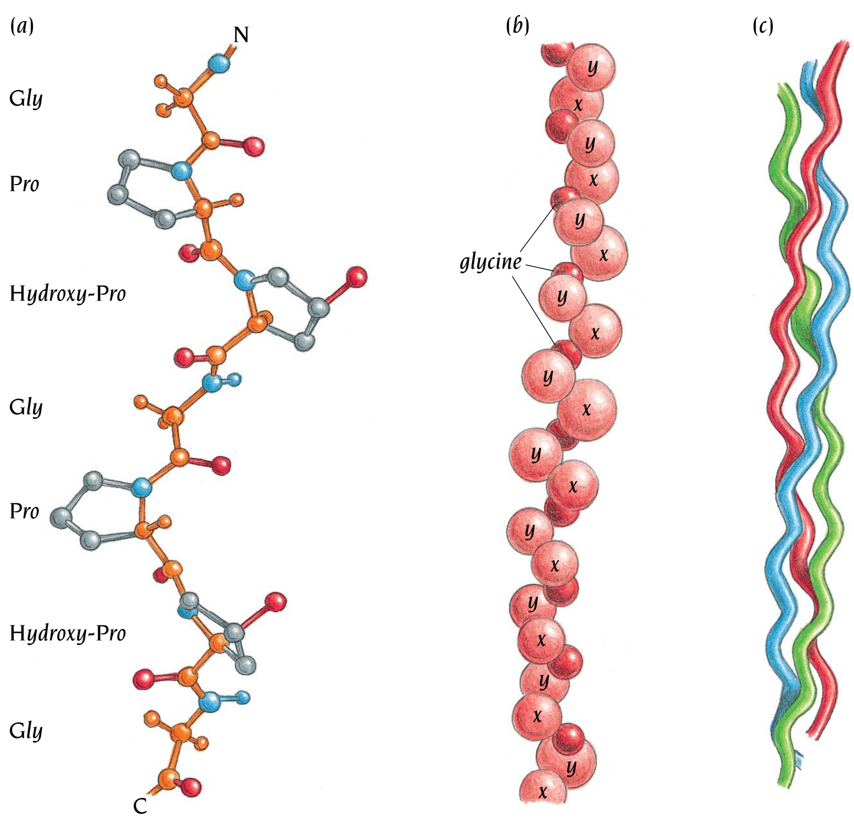


Figure 14.1 Each polypeptide chain in the collagen molecule folds into an extended polyproline type II helix with a rise per turn along the helix of 9.6 Å comprising 3.3 residues. In the collagen molecule three such chains are supercoiled about a common axis to form a 3000-Å-long rod-like molecule. The amino acid sequence contains repeats of -Gly-X-Y- where X is often proline and Y is often hydroxyproline. (a) Ball and stick model of two turns of one polypeptide chain. (b) A model of one collagen chain in which each amino acid is represented by a sphere. (c) A small part of the collagen superhelix in which the three chains have different color. [(a) Adapted from R. Fraser et al., *J. Mol. Biol.* 129: 463–481, 1979. (b) Adapted from B. Alberts et al., *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.]

chains: they form interchain disulfide bonds that align the three chains in register before formation of the triple chain structure. Only following exocytosis are the propeptides cleaved off, by extracellular enzymes. Excision of both propeptides allows the triple chain molecules to polymerize into fibrils several micrometers long and 50–200 nanometers in diameter. These fibrils then pack side by side into parallel bundles, the collagen fibers, which are stronger than steel of the same size. When mature collagen is denatured, since there are no propeptides present, the polypeptides associate at many different places other than their ends, forming triple chains that are out of register. These polymerize to form a gel, gelatin.

Early interpretations of fiber diffraction studies of collagen by, among others, Linus Pauling, Francis Crick, and Alexander Rich, established that each of the three polypeptide chains is folded into an extended left-handed helix with 3.3 residues per turn and a rise per residue along the helical axis of 2.9 Å. In contrast the more compact right-handed α helix has 3.6 residues per turn and a rise per residue of 1.5 Å (see Chapter 2). The rise per turn in the collagen helix is therefore 9.6 Å, compared with 5.4 Å for the α helix, and this gives such an extended chain that it must aggregate to form a stable structure (Figure 14.1). Synthetic polymers of proline or glycine fold into similar extended, left-handed helices and so this helix type is called a polyproline type II helix. SH3 domains, described in Chapter 13, recognize short proline-rich segments in other proteins and these segments have a similar helical conformation when bound to SH3.

The three polyproline type II helices in collagen form a trimeric molecule by coiling about a central axis to form a right-handed superhelix with a repeat distance of about 100 Å (see Figure 14.1c). In this superhelix the side chain of every third residue is close to the central axis where there is no space available for a side chain, consequently every third residue must be a glycine. Any other residue deforms the superhelix and certain inherited connective tissue diseases are due to mutations in codons for these glycine residues. This sequence requirement is a hallmark of triple helix collagen-like domains that is used in sequence analyses of proteins of unknown structure. Triple-helical domains are found in a variety of supramolecular aggregates, ranging from the collagen fibrils in tendons and cartilage, to network forms seen in basement membranes, and to the parallel clusters of short triple helices seen in the blood complement component C1q or in the sugar-binding collectins.

A detailed picture of the collagen triple helix has been obtained by the group of Helen Berman at Rutgers University, who have determined the crystal structure to 1.9 Å resolution of a synthetic collagen-like peptide (Pro-Hydroxypro-Gly)$_{10}$ with one Gly substituted by Ala. This collagen-like peptide did not form fibers but single crystals. The specific sequence was chosen in order to study both the details of the regular collagen triple helix structure and the effect of mutation at a glycine position. The structure shows the importance of direct as well as water-mediated hydrogen bonds in stabilizing the triple helix structure. In addition it shows that the alanine side chain can be accommodated inside the triple helix by a local small change of the helix geometry (Figure 14.2), which allows the incorporation of interstitial water molecules to link the chains. Such conformational shifts may help to accommodate sequence variations that deviate from the consensus.

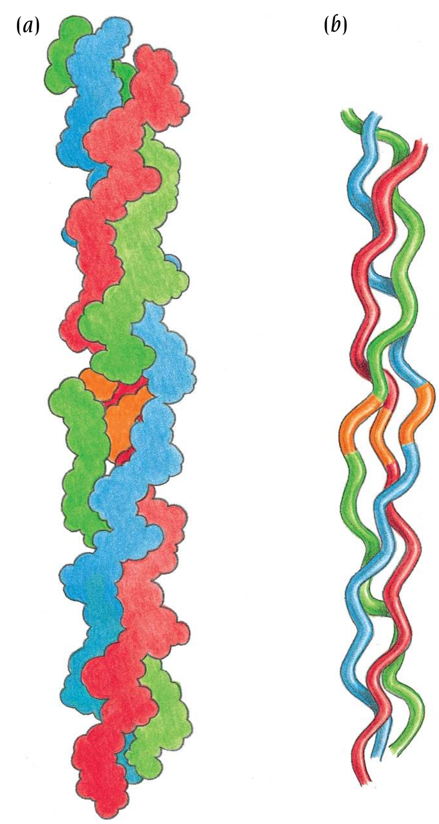
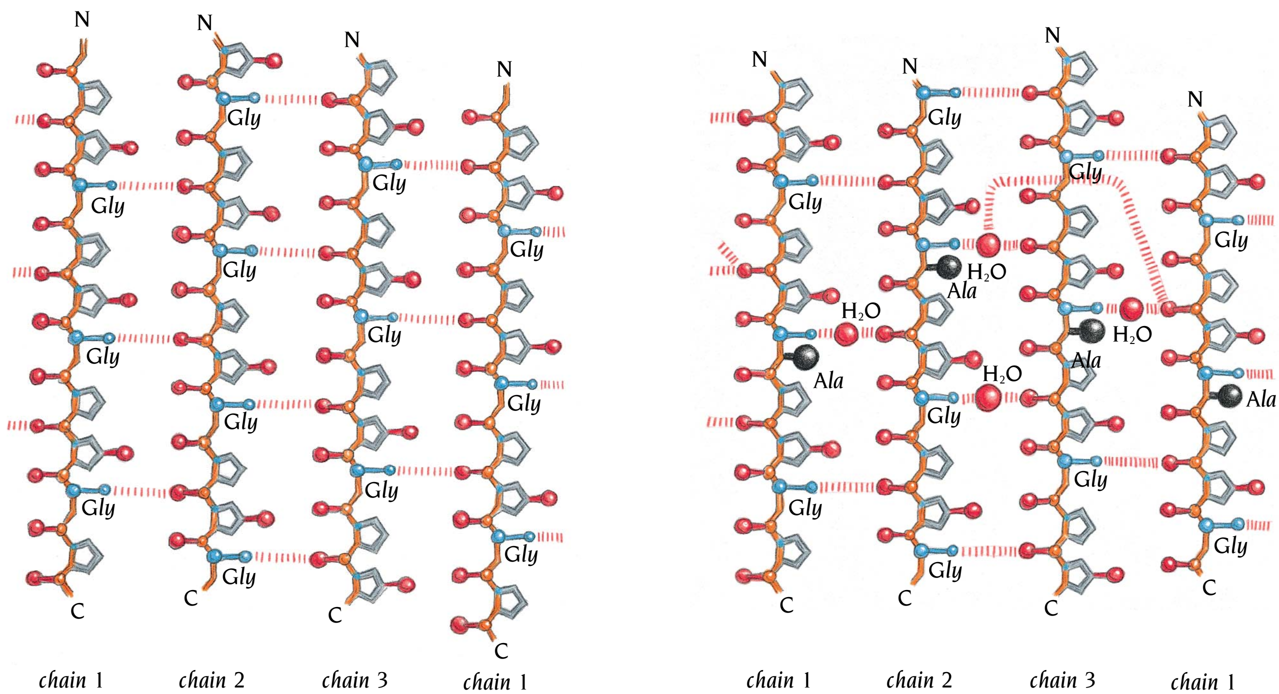


Figure 14.2 Models of a collagen-like peptide with a mutation Gly to Ala in the middle of the peptide (orange). Each polypeptide chain is folded into a polyproline type II helix and three chains form a superhelix similar to part of the collagen molecule. The alanine side chain is accommodated inside the superhelix causing a slight change in the twist of the individual chains. (a) Space-filling model. (b) Ribbon diagram. Compare with Figure 14.1c for the change caused by the alanine substitution. (Adapted from J. Bella et al., *Science* 266: 75–81, 1994.)



In the regular, triple helix collagen structure the three chains are held close together by direct hydrogen bonds between proline C=O groups of one chain and the glycine NH groups of another (Figure 14.3a). In the region around the alanine residues the three polypeptide chains are forced apart by the alanine side chains and four water molecules are incorporated between the chains; this allows the direct hydrogen bonds to be replaced by water-mediated hydrogen bonds (Figure 14.3b).

All side chains as well as the C=O group of glycines in all three chains are on the outside of the triple helix molecule and in contact with water molecules. These water molecules mediate hydrogen bonds between the hydroxyl groups of hydroxyproline and the peptide C=O and NH groups both within each chain and between different chains (Figure 14.4). These water mediated hydrogen bonds are essential for the stability of the triple helix and are presumably the reason for the presence of hydroxyproline in collagen.

Figure 14.3 Schematic diagrams of interchain hydrogen bonding (stripes) in the collagen triple helix showing a cylindrical projection with chain 1 repeated at the right side to provide a clear picture of the chain 3 to chain 1 hydrogen bonds. The three chains are staggered by one residue and go clockwise from 1 to 3. (a) Interchain hydrogen bonds in regular collagen triple helix between proline C=O groups and glycine NH groups. (b) Water-mediated hydrogen bonds in the middle of the Gly-Ala substituted peptide structure. Four water molecules are inserted in the interior of the triple helix to mediate hydrogen bonds between the polypeptide chains, which are displaced due to the alanine side chains in this region. Interchain hydrogen bonds are also shown. (Adapted from J. Bella et al., *Science* 266: 75–81, 1994.)

Coiled coils are frequently used to form oligomers of fibrous and globular proteins

We described in Chapter 3 the basic features of α-helical coiled coils whose amino acid sequences are recognized by heptad repeats *a* to *g* in which positions *a* and *d* frequently are hydrophobic residues (see Figures 3.2 and 3.3).

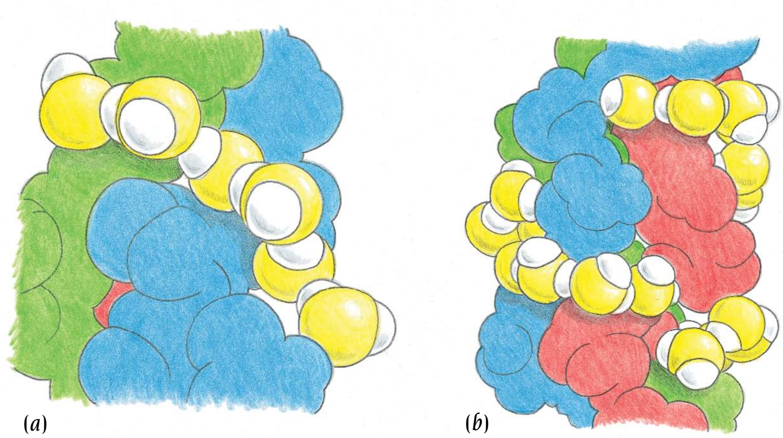


Figure 14.4 Space-filling models of intrachain water bridges observed in the crystal structure of a collagen-like peptide. The three peptide chains are green, blue and red. The oxygen atoms of the water molecules are yellow and the hydrogen atoms are white. (a) A bridge of three water molecules hydrogen bonded to two C=O groups and linked to other water molecules. (b) Long bridges of water molecules establish a network that surrounds the triple helix. (Adapted from J. Bella et al., *Structure* 3: 893–906, 1995.)

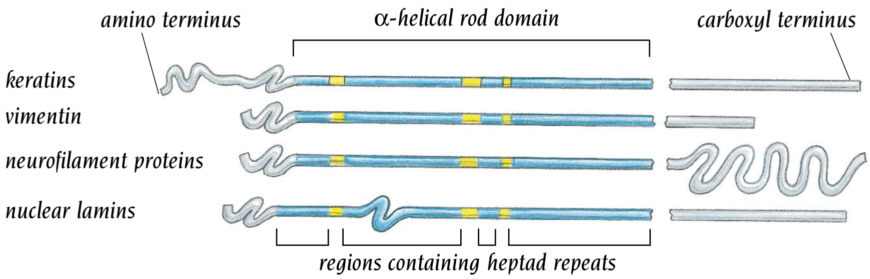


Figure 14.5 The domain organization of intermediate filament protein monomers. Most intermediate filament proteins share a similar rod domain that is usually about 310 amino acids long and forms an extended α helix. The amino-terminal and carboxy-terminal domains are non-α-helical and vary greatly in size and sequence in different intermediate filaments. (Adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)

The leucine zipper DNA-binding proteins, described in Chapter 10, are examples of globular proteins that use coiled coils to form both homo- and heterodimers. A variety of fibrous proteins also have heptad repeats in their sequences and use coiled coils to form oligomers, mainly dimers and trimers. Among these are myosin, fibrinogen, actin cross-linking proteins such as spectrin and dystrophin as well as the intermediate filament proteins keratin, vimentin, desmin, and neurofilament proteins.

Keratins, which are the major structural protein in skin, hair, and feathers, belong to the superfamily of **intermediate filaments** that comprise about 60 genes of greatly differing sizes in the human genome. All members of this family contain a homologous central region of about 300 amino acid residues that exhibit heptad repeats typical for coiled coil formation (Figure 14.5). It is assumed that these central regions form a double-stranded coiled coil giving a homodimer with a rod-like central domain and two globular domains, one at each end. In assembling into an intermediate filament, the rod-like domains interact with one another to form the uniform core of the filament, while the globular domains, which vary in size in different intermediate filament proteins, project from the filament surface. A hypothetical view of the assembly of the peptide chains into intermediate filaments is shown in Figure 14.6.

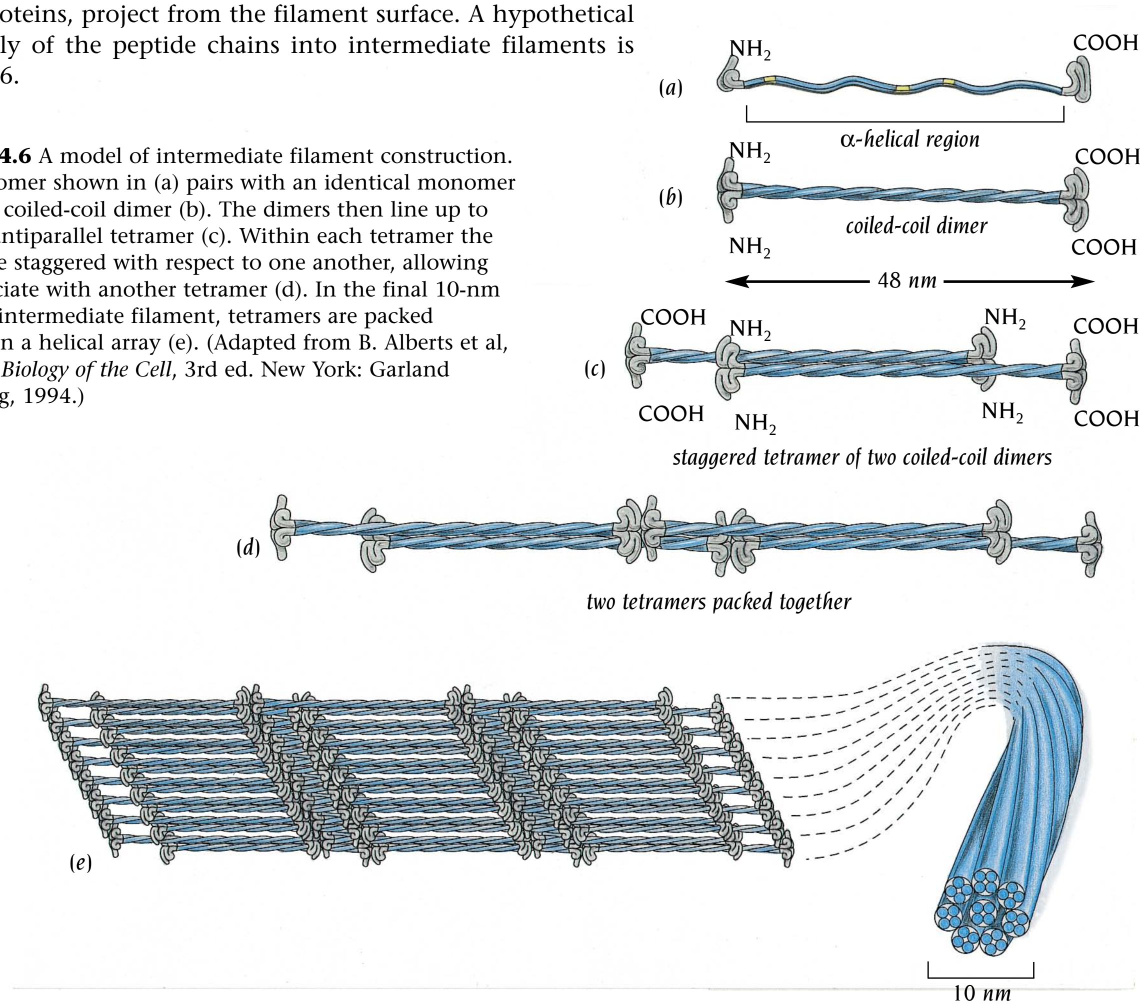


Figure 14.6 A model of intermediate filament construction. The monomer shown in (a) pairs with an identical monomer to form a coiled-coil dimer (b). The dimers then line up to form an antiparallel tetramer (c). Within each tetramer the dimers are staggered with respect to one another, allowing it to associate with another tetramer (d). In the final 10-nm rope-like intermediate filament, tetramers are packed together in a helical array (e). (Adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)

Amyloid fibrils are suggested to be built up from continuous β sheet helices

Amyloidosis are diseases in which proteins that are normally globular and soluble are deposited as stable, insoluble fibrils in the extracellular space of tissues such as the brain or the eye. Deposition of such fibrils is associated with several amyloid diseases including the transmissable spongiform encephalopathies (BSE and Creutzfeld–Jacob diseases), Alzheimer's disease and type II diabetes. **Amyloid fibrils** from different sources frequently share a common ultrastructure; they are long, straight, unbranched rods about 100 Å in diameter. A unifying model for fibril structures has been proposed on the basis of amyloid fibers of the protein transthyretin. Colin Blake and Louise Serpell at Oxford University have obtained fiber diffraction patterns to high resolution, using synchrotron radiation, of such amyloid fibers from patients suffering from familial amyloidotic polyneuropathy. These patients have an inherited single-point mutation, Val to Met, in the protein **transthyretin** which carries the hormone thyroxine in the bloodstream. This seemingly innocuous mutation causes the soluble, globular transthyretin molecule to polymerize and gradually deposit as fibrous material in the heart or eye. The group of Colin Blake had earlier determined the crystal structure of the soluble form of transthyretin and showed it to be a homotetramer with the subunits folded into an antiparallel β structure (Figure 14.7).

Earlier fiber diffraction studies to lower resolution of amyloid fibers had established that they have a β structure, with fibers from different sources giving similar diffraction patterns. Structural models had been suggested based on planar pleated β sheets, similar to a model suggested for silk fibers, which we shall discuss later. Blake and Serpell's high resolution data, however, showed a distinct, repeating pattern corresponding to structural repeats of 115.5 Å along the fiber axis. Because almost all β sheets in globular proteins are twisted and not planar, they proposed, therefore, that the 115.5 Å repeats they observed are compatible with a twisted β sheet completing one revolution of 360° every 115.5 Å (Figure 14.8). Since the mean distance between β strands in globular proteins is about 4.8 Å, the repeating unit along the fibril would correspond to 24 β strands with an average twist of 15° between each successive β strand, a value that is in good agreement with the values observed in β sheets in globular proteins. Finally, they suggested that each repeat unit is built up from four transthyretin molecules, each contributing six β strands to the β sheet helix. The model includes the idea that a partially unfolded form that has lost two strands forms the fibril. This idea is supported by the absence of fibrils in the brain, where hormone binding stabilizes the native state.

Electron microscopy of transthyretin fibrils shows that they have a uniform width of about 130 Å and are composed of four parallel protofilaments each 50–60 Å in diameter. Blake and Serpell propose that each protofilament is built up of four continuous twisted β sheets of indefinite length running parallel to the axis of the fibril, with their constituent β strands, each containing about 10 amino acid residues, running perpendicular to the axis. The four β sheets are arranged in pairs, each pair being stabilized by a hydrophobic core between the two sheets, as is usually the case in globular proteins with β structure. The two pairs of β helices coil about each other. The twisted β helix only comprises the regular, diffracting, core structure of the protofilament and must be surrounded by other material such as loops of polypeptide chains that do not contribute to the x-ray pattern.

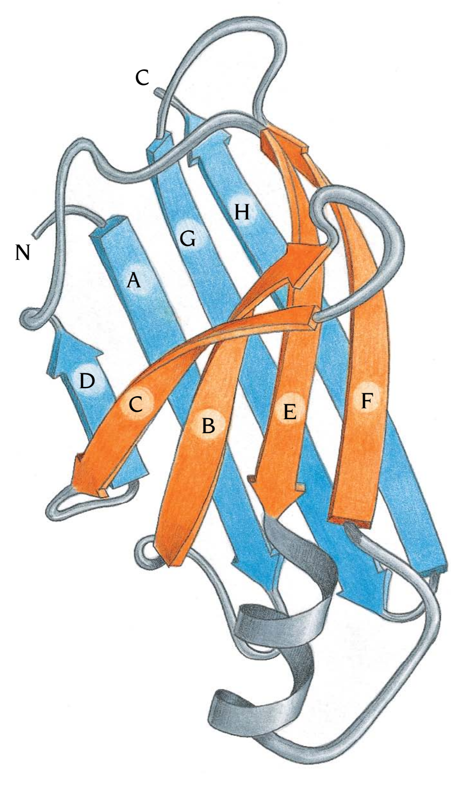
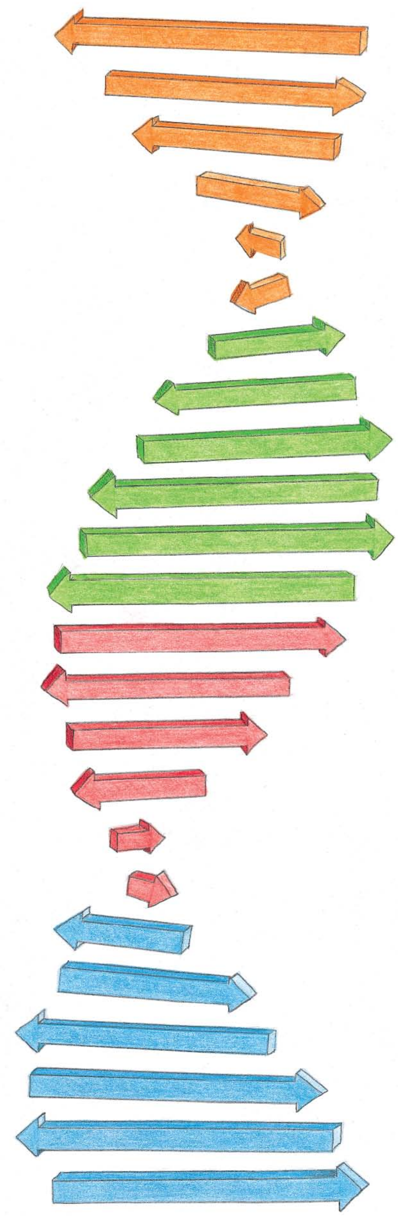


Figure 14.7 Ribbon diagram of one subunit of the globular form of transthyretin. The β strands are labeled A to H from the amino end. Strands C and D are thought to be unfolded to produce the conformation that forms amyloid fibrils. (Adapted from C.C.F. Blake et al., *J. Mol. Biol.* 121: 339–356, 1978.)

Figure 14.8 Proposed model of β sheet helix of the fibrous form of transthyretin. The repeating unit of the β helix comprises 24 β strands with an average twist of 15° between each strand giving a complete turn of 360°. Four transthyretin polypeptide chains contribute to the repeat unit and are shown here in different colors. (Adapted from C. Blake and L. Serpell, *Structure* 4: 989–998, 1996.)



This model implies a considerable refolding of the globular state of transthyretin to form the fibrous state. However, large refolding events are known to occur in other proteins which form amyloid β sheet fibers from subunits that are mainly α-helical in their globular state, for example prions and α lactalbumin. Even though not every detail of this model may be correct, it not only is compatible with the fiber diffraction model but also incorporates knowledge gained from structural studies of β sheets in globular proteins.

Spider silk is nature's high-performance fiber

Individual spiders generate up to seven distinct silk fibers by drawing liquid-crystalline proteins from a set of separate gland-spinneret complexes. The proteins, called **silk fibroins**, are produced inside the glands from a family of homologous genes and stored as a highly concentrated (up to 50%) solution of mainly α-helical globular molecules. This solution is passed through the spinning machinery of the spider and mixed with other components, producing a variety of different fibers in which the proteins have adopted a β structure. Spiders can produce different types of silk with different mechanical properties for different functions. The most remarkable properties are exhibited by the dragline fiber that the spider spins rapidly and uses to climb from the web to its hiding place. This fiber is stronger than steel, yet it can contract, stretch out and bend without breaking. The structure of silk fibers has therefore attracted the attention of materials scientists who are increasingly willing to turn to biology for lessons that might be applied to synthetic systems.

Several silk fibroin genes have been cloned and sequenced and they all show a similar sequence pattern: variable domains at the N- and C-termini flank a large region of repetitive short sequences of alternating poly-Ala (8 to 10 residues) and Gly-Gly-X repeats (where X is usually Ser, Tyr, or Gln). This middle region varies in length and may comprise up to 800 residues.

Early fiber diffraction studies in the 1950s established that at least some of the proteins in the spider silk fibers adopt a β sheet structure, similar to the β sheets that had been observed in common silk. Simple models of planar β sheets were suggested to explain these diffraction patterns. They were, however, at very low resolution due to the difficulty in aligning thousands of fibers into a large enough bundle to obtain visible diffraction spots. Bundles of fibers that are not precisely aligned give broad and diffuse diffraction patterns that do not reveal structural details (see Chapter 18). Not until very recently has it been possible to obtain x-ray diffraction patterns from individual spider dragline fibers that are only a few micrometers thick, using the very intense x-ray beams from the new generation of synchrotrons.

Progress in deducing more structural details of these fibers has instead been achieved using NMR, electron microscopy and electron diffraction. These studies reveal that the fibers contain small microcrystals of ordered regions of the polypeptide chains interspersed in a matrix of less ordered or disordered regions of the chains (Figure 14.9). The microcrystals comprise about 30% of the protein in the fibers, are arranged in β sheets, are 70 to 100 nanometers in size, and contain trace amounts of calcium ions. It is not yet established if the β sheets are planar or twisted as proposed for the amyloid fibril discussed in the previous section.

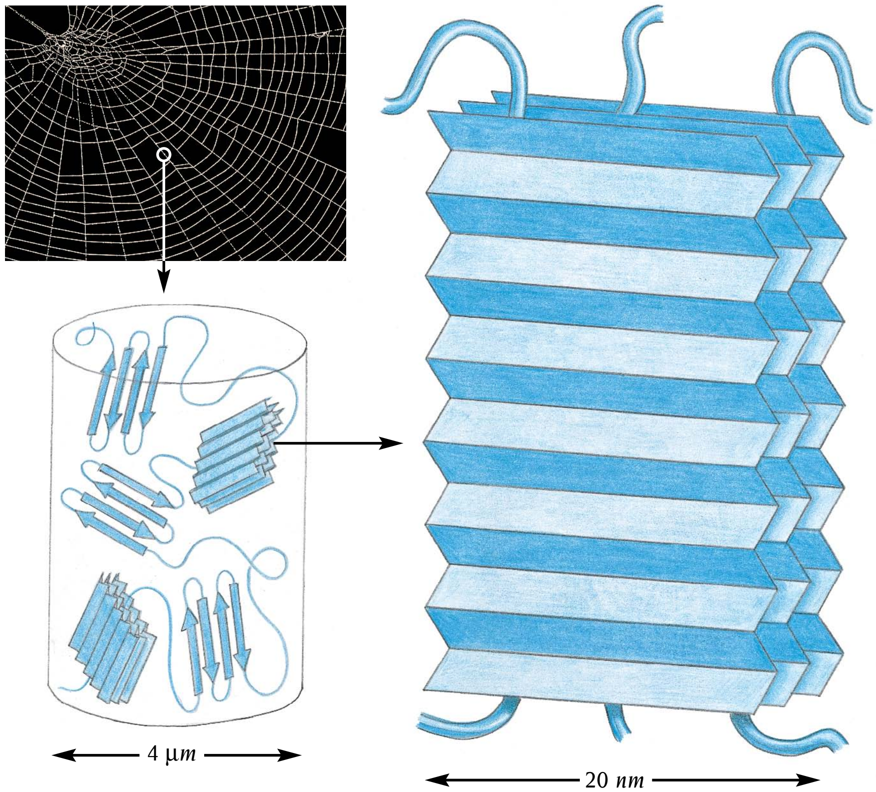


Figure 14.9 Spider fibers are composite materials formed by large silk fibroin polypeptide chains with repetitive sequences that form β sheets. Some regions of the chains participate in forming 100-nm crystals, while other regions are part of a less-ordered meshwork in which the crystals are embedded. The diagram shows a model of the current concepts of how these fibers are built up, which probably will be modified and extended as new knowledge is gained. (Adapted from F. Vollrath, *Sci. Am.* p. 54–58, March 1992 and A.H. Simmons, *Science* 271: 84–87, 1996. Photograph courtesy of Science Photo Library.)

An elegant NMR experiment by the group of Lynn Jelinski at Cornell University has established that at least part of the microcrystals is built up from the polyalanine repeats in the protein chains. These experiments, which were made on ^{13}C-enriched proteins produced by feeding the spiders ^{13}C-labeled alanine, showed that there were two populations of alanine side chains, one ordered and oriented perpendicular to the fiber axis and a second less ordered. Jelinski's interpretation is that parts of the polyalanine sequences are incorporated as β strands in the microcrystals with an orientation parallel to the fiber axis. Whether or not the Gly-Gly-X repeats also form β strands in the microcrystals remains an open question.

Even if much remains to be done to clarify the relations between structure and mechanical properties of the spider fibers, the knowledge obtained so far gives two important messages. First, the process of spinning converts the silk proteins from a soluble α-helical form to a fibrous form containing β sheets, analogous to the conversion in humans and other animals of soluble amyloid and prion proteins to insoluble fibrous forms. Second, the mechanical properties of silk fibers, like those of synthetic fibers, are dependent on having well-ordered crystalline regions which give strength, within a matrix of less-ordered regions, which give flexibility. An understanding of the structural details of these regions, as well as the interplay between them, might pave the way for fascinating new materials.

Muscle fibers contain myosin and actin which slide against each other during muscle contraction

Muscle contraction takes place by the mutual sliding of two sets of interdigitating filaments made of fibrous proteins: **thick filaments** (containing **myosin**) and **thin filaments** (containing **actin**). The thick and thin filaments are organized in basic contractile units called sarcomeres, each 2–3 μm long, which also contain other proteins and give muscle its cross-striated appearance in the microscope (Figure 14.10). Another fibrous component of the sarcomere, titin, is the largest known polypeptide chain, with a molecular

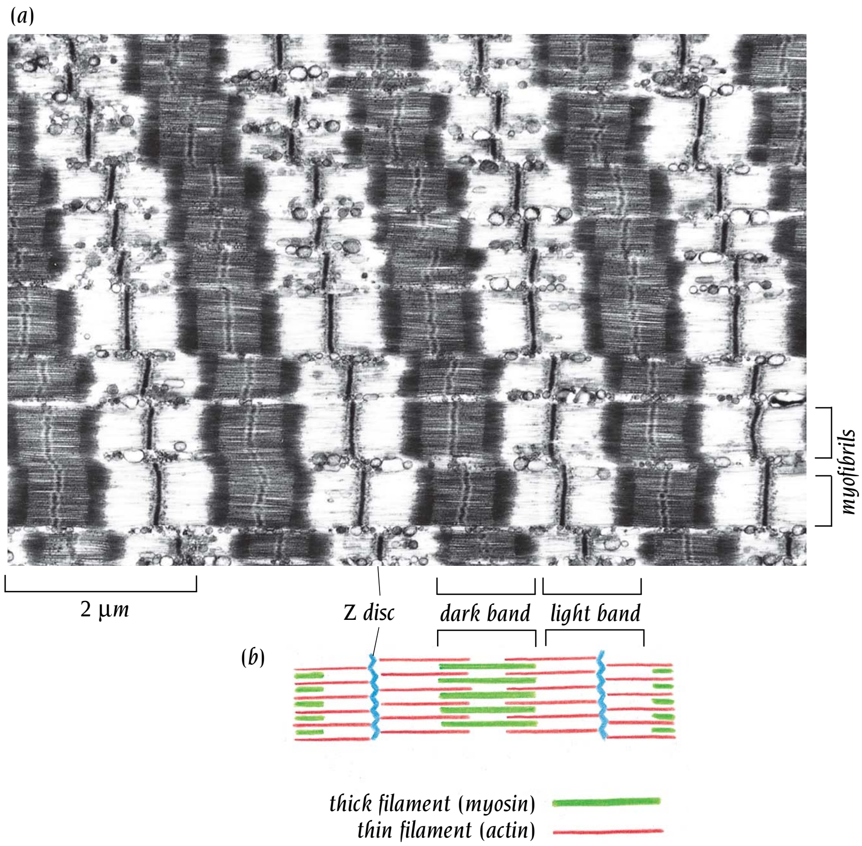


Figure 14.10 A muscle viewed under the microscope is seen to contain many myofibrils that show a cross-striated appearance of alternating light and dark bands, arranged in repeating units called sarcomeres. The dark bands comprise myosin filaments and are interupted by M (middle) lines, which link adjacent myosin filaments to each other. The light bands comprise actin filaments that are attached to disks at each end of the sarcomeres. (Courtesy of Roger Craig, adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)

weight of approximately 3000 kDa. Titin is thought to help measure the length of the sarcomere and return the stretched muscle to the correct length. Figure 14.11 shows how the sarcomeres contract as the actin and myosin filaments slide past each other.

Myosin heads form cross-bridges between the actin and myosin filaments

Within each sarcomere the relative sliding of thick and thin filaments is brought about by "**cross-bridges**," parts of the myosin molecules that stick out from the myosin filaments and interact cyclically with the thin actin filaments, transporting them by a kind of rowing action. During this process, the hydrolysis of ATP to ADP and phosphate couples the conformational

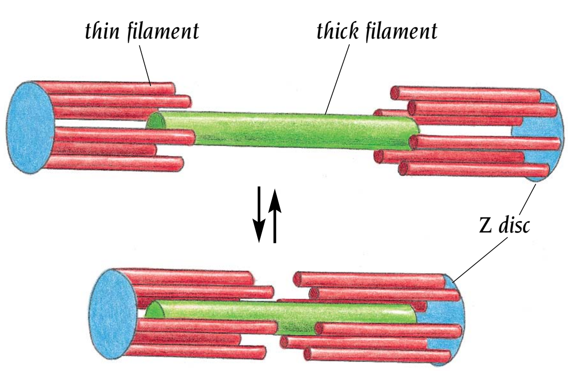


Figure 14.11 The sliding filament model of muscle contraction. The actin (red) and myosin (green) filaments slide past each other without shortening. (Adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)

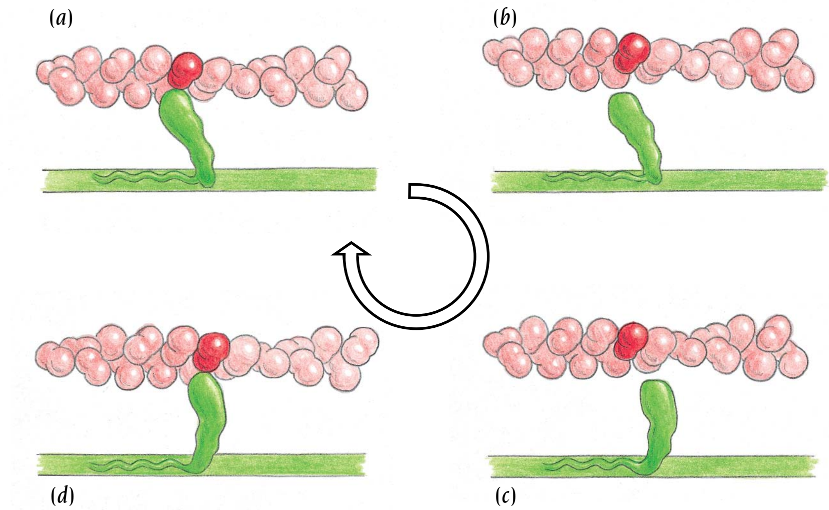


Figure 14.12 The swinging cross-bridge model of muscle contraction driven by ATP hydrolysis. (a) A myosin cross-bridge (green) binds tightly in a 45° conformation to actin (red). (b) The myosin cross-bridge is released from the actin and undergoes a conformational change to a 90° conformation (c), which then rebinds to actin (d). The myosin cross-bridge then reverts back to its 45° conformation (a), causing the actin and myosin filaments to slide past each other. This whole cycle is then repeated.

changes in myosin to actin binding and release. Myosin in the thick filaments is a fibrous protein with individual chains arranged in helical coiled coils. Actin by contrast is a fibrous protein formed by linking together globular monomeric subunits.The first molecular theories of muscle contraction, which appeared in the 1930s, were based on polymer science. They proposed that there was a rubber-like shortening of myosin filaments brought about by altering the state of ionization of myosin. This aberration was corrected in 1954 by the work of H.E. and A.F. Huxley, which showed that sarcomeres contain two sets of filaments that glide over each other without altering their length. The question naturally arose: what makes them glide?

Following the discovery by electron microscopy of the myosin cross-bridges to the actin filaments, two conformations of the cross-bridge were observed in insect flight muscle. This seminal finding led to the **swinging cross-bridge model** to explain the sliding of the actin filaments into myosin filaments (Figure 14.12). The myosin cross-bridge was thought to bind to actin in an initial (90°) conformation and then go over to an angled (45°) conformation, followed by release from the actin. For each complete cycle one molecule of ATP would be hydrolyzed. The actual movement per cycle of ATP hydrolysis was measured in physiological experiments to be about 80–100 Å and since the cross-bridge was an elongated structure such a movement could be accommodated by swinging the cross-bridge. However, this model basically rested on the observation of two conformations of cross-bridges in fixed and sectioned insect flight muscle under the electron microscope. It required the development of synchrotron radiation x-ray beams to provide more direct evidence.

Time-resolved x-ray diffraction of frog muscle confirmed movement of the cross-bridges

Excised living frog muscles will continue to contract for many hours if bathed in Ringer solution and stimulated electrically. H.E. Huxley showed that living frog muscles give detailed low-angle x-ray fiber diffraction patterns (see Chapter 18) with a series of layer-lines arising from the helical arrangement of cross-bridges. A strong reflection on one of these layer-lines indicates that the distance between cross-bridges along the myosin filament helix is 143.5 Å. If the cross-bridges tilt during contraction, then this reflexion should get weaker. Muscles contract fast, so millisecond time resolution is necessary. Initial attempts to observe this effect with conventional x-ray sources failed: the signal in the short times available for the diffractions was too weak. The first beam lines for x-ray diffraction experiments using synchrotron radiation were set up at DESY, Hamburg, by Ken Holmes and his colleagues from Heidelberg; these provided the requisite power to allow data collection over only a few milliseconds. The key experiments, carried out in 1980 by H.E. Huxley, showed the anticipated changes in intensity of the diffraction data. If a contracting muscle is allowed to relax, the intensity of the strong reflection arising from the cross-bridges drops within a few milliseconds to a fraction of its initial value. If one waits some time with the muscle at its new length, the intensity recovers. These and subsequent time-resolved experiments at synchrotrons are fully consistent with the swinging cross-bridge hypothesis.

Structures of actin and myosin have been determined

Structural studies to atomic resolution of both actin and myosin have clarified a number of long-standing questions concerning the mechanism of cross-bridge movements in muscle. Fibrous actin, F-actin, is a helical polymer of a globular polypeptide chain, G-actin, comprising about 375 amino acid residues. The first crystal structure of a monomeric G-actin molecule was determined by the group of Ken Holmes, Max-Planck Institute, Heidelberg, who prevented polymerization by forming a complex with the enzyme DNAse. Subsequently the structures of several other G-actin molecules have been determined in complex with other inhibitors of actin polymerization. The structure comprises four domains, two of which are similar α/β domains that contain an ATPase catalytic site (Figure 14.13). Related structures have also been found in such diverse ATP-requiring enzymes as hexokinase and the chaperone hsc70. These molecules probably form an evolutionary family, but their specific functions and amino acid sequences have completely diverged. Only their ATPase activity and three-dimensional structures are retained.

The F-actin helix has 13 molecules of G-actin in six turns of the helix, repeating every 360 Å. Oriented gels of actin fibers yield x-ray fiber diffraction patterns to about 6 Å resolution. Knowing the atomic structure of G-actin it was possible for the group of Ken Holmes to determine its orientation in the F-actin fiber, and thus arrive at an atomic model of the actin filament that best accounted for the fiber diffraction pattern.

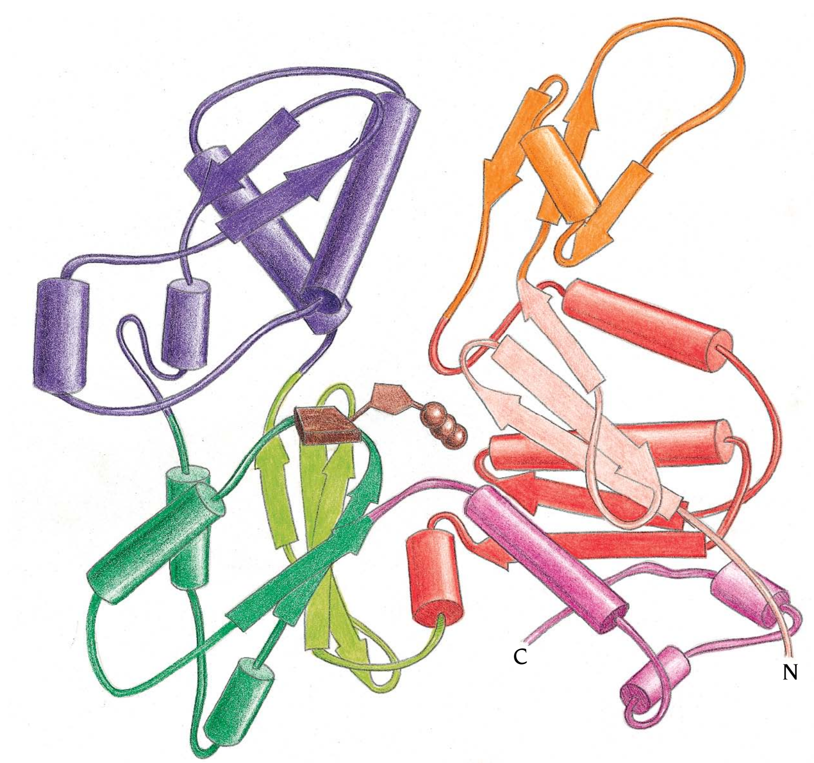


Figure 14.13 Structure of G-actin. Two α/β-domains, (dark and light red and dark and light green) bind an ATP molecule (brown) between them. This ATP is hydrolyzed when the actin monomer polymerizes to F-actin.

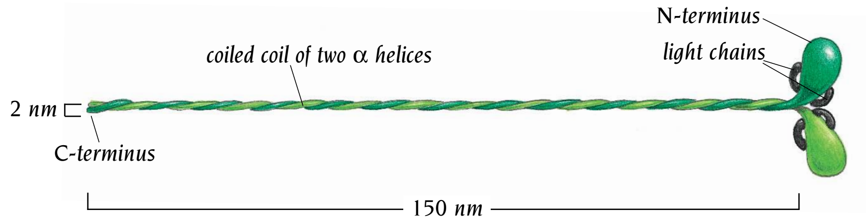


Figure 14.14 Schematic diagram of the myosin molecule, comprising two heavy chains (green) that form a coiled-coil tail with two globular heads and four light chains (gray) of two slightly differing sizes, each one bound to each heavy-chain globular head.

The myosin molecule, which is a dimer consisting of two heavy chains and four light chains, forms a 1400-Å-long tail and two heads, each of 120,000 kDa molecular weight (Figure 14.14). The C-terminal regions of the heavy chains are folded into long α helices that form the tail by dimerizing through parallel coiled coils. Fragments of myosin called subfragment 1, or S1, can be cleaved off; S1 consists of the two light chains and the N-terminal region of one heavy chain that includes the globular head and a short part of the helical tail. The structure of S1 from chicken myosin has been determined by the group of Ivan Rayment at the University of Wisconsin, Madison. The fragment is tadpole-like in form, with an elongated head and a tail (Figure 14.15). The head contains a seven-stranded β sheet (green) and numerous associated α helices (red) which form a deep cleft, with the actin-binding site and the nucleotide-binding site on opposite sides of the β sheet. The ATP-binding site contains a typical switch II region, which is also found in the G-proteins (see Chapter 13) that changes conformation depending on whether ATP or ADP is bound. The cleft separates two parts of the head, which are referred to as the 50K upper and 50K lower domains (see Figure 14.15b). The tail, which also provides the connection to the thick filament, has a long extended α helix formed by the heavy chain that binds two light chains (yellow and magenta in Figure 14.15).

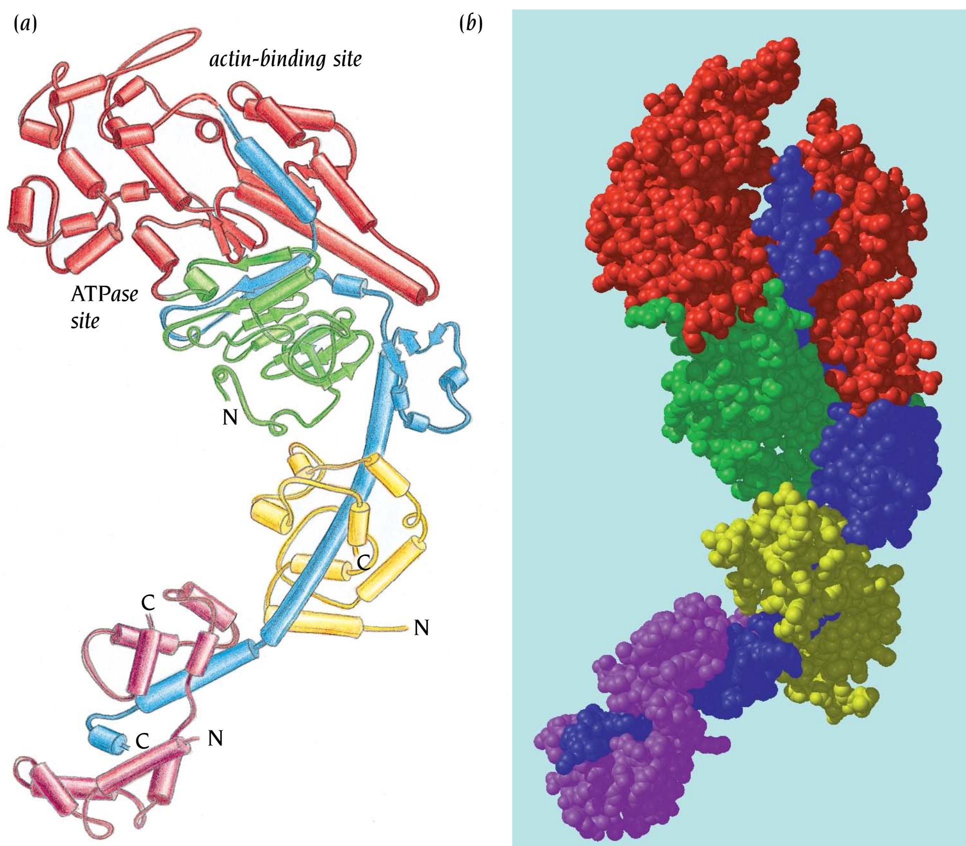


Figure 14.15 Structure of the S1 fragment of chicken myosin as a Richardson diagram (a) and a space-filling model (b). The two light chains are shown in magenta and yellow. The heavy chain is colored according to three proteolytic fragments produced by trypsin: a 25-kDa N-terminal domain (green); a central 50-kDa fragment (red) divided by a cleft into a 50K upper and a 50K lower domain; and a 20-kDa C-terminal domain (blue) that links the myosin head to the coiled-coil tail. The 50-kDa and 20-kDa domains both bind actin, while the 25-kDa domain binds ATP. [(b) Courtesy of I. Rayment.]